Learning From Itself Without Losing Itself: A Label-Free Defense Against Model Collapse in LLMs

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Abstract

Premise. High-quality human text is becoming a finite bottleneck for scaling language-model training, pushing the field toward **synthetic data**. Yet naïve recursive self-feeding induces **model collapse**: rare tokens disappear, outputs converge to a stylistic monoculture, label noise compounds, and rare knowledge decays first.

We propose the **Anti-Collapse Protocol (ACP)**, a **provenance-agnostic, label-free** recipe for repeated synthetic fine-tuning that preserves *heterozygosity* without detecting "AI vs. human." ACP couples a **label-free consensus gate**, which admits an answer only when independent reasoning routes agree, with four information-geometry gates: **(T1) Entropy Banding** admits learnable, mid-surprisal samples; **(T2) Tail-Mass & Novelty Floors** with near-duplicate caps preserve rare-token throughput; **(T3) Domain Isotropy** prevents topical lock-in; and **(T4) Red-Queen Critics** over a route lattice preserve multiple solution mechanisms. ACP runs as a closed loop, **Synthesize** $\rightarrow$ **Screen** $\rightarrow$ **Train** $\rightarrow$ **Measure** $\rightarrow$ **Adapt**, with tripwires.

We evaluate ACP in a **fully self-consuming loop**: four model families $\times$ four benchmarks $\times$ eight generations, with **zero gold labels**. Accuracy stays within noise of the base model in 14 of 16 settings (mean -0.72 pp; none significant after correction). The student's own outputs erode, yet admitted-label precision exceeds generated precision in **all 128 rounds**, cutting label noise from 20.0% to 11.0%, with the largest gains where the generator is weakest ($r=-0.75$). An unfiltered loop matches ACP's accuracy while training on **48% wrong labels**: accuracy is a *lagging* indicator of collapse. A consensus-precision bound shows that route diversity decorrelates *reasoning* errors but not *knowledge* errors. On open web text with four backbones, ACP raises rare-token mass by 13–31% while retaining $\geq$97.6% of teacher–route diversity.

The Anti-Collapse Protocol

- ***Information-geometry framing:*** Collapse is a label, entropy, tail, novelty and mechanism failure—not a provenance failure. ACP regulates admitted label precision, entropy bands, tail mass, novelty, isotropy and route plurality rather than "AI vs. human" origin.
- **ACP *core—one consensus gate + four constraints:*** a closed loop **Synthesize** $\rightarrow$ **Screen (consensus, T1–T4)** $\rightarrow$ **Train** $\rightarrow$ **Measure** $\rightarrow$ **Adapt**. No gate reads a gold label or a provenance tag.
- ***Compute-feasible instantiation:*** 7–9B LoRA/QLoRA students. **Regime I**: Gemma-2-9B, Llama-3.1-8B, Mistral-7B, Qwen2.5-7B each generate their *own* data for GSM8K, ARC-C, CSQA, BoolQ for 8 generations. **Regime II**: a two-teacher lattice on FineWeb, 4 backbones $\times$ 8 rounds.
- ***Diagnostics & tripwires:*** label precision, H_{π_0}-band drift, Zipf-tail mass, Heaps β, size-matched repetition, route entropy, teacher–route MI, counterfactual catch rate.
- ***Mechanism diversity:*** five routes (direct, paraphrase-first, step-by-step, contrastive, Socratic) kept plural by route caps, route-aware sampling and rotating critics.
- ***Empirical stability under recursion:*** accuracy within noise in 14/16 settings; admitted data purer than generated data in 128/128 rounds; label noise 20.0%$\rightarrow$11.0%.
- ***Accuracy lags collapse:*** an unfiltered loop reaches the same accuracy while 48% of its training labels are wrong (16.5$\times$ ACP).
- ***Self-audit:*** a consensus bound measures route-error correlation; a size audit removes 85% of our own apparent repetition gain; gates that never bound are reported as such.
- ***Reviewer-auditable protocol:*** every number and figure is regenerated from released per-round logs.

1 The Anti-Collapse Protocol: Necessity & Urgency

Data-exhaustion premise. Multiple independent analyses suggest that the ***quality-adjusted stock of public, human-authored web text is finite*** and may be substantially consumed within the next few frontier training cycles. Forecasts combining historical dataset growth with compute-scaled training trends estimate an accessible stock of roughly $4\text{–}5 \times 10^{14}$ tokens and project that frontier-scale runs may reach it by the late 2020s (Villalobos et al., 2024). Compute-optimal scaling ties capability gains to both parameters and training tokens, structurally increasing demand for fresh data (Kaplan et al., 2020; Hoffmann et al., 2022). Future progress therefore depends on procedures that can **(i) generate new signal, (ii) curate it for quality and diversity, and (iii) stabilize learning under repeated reuse** (Fig. 1).

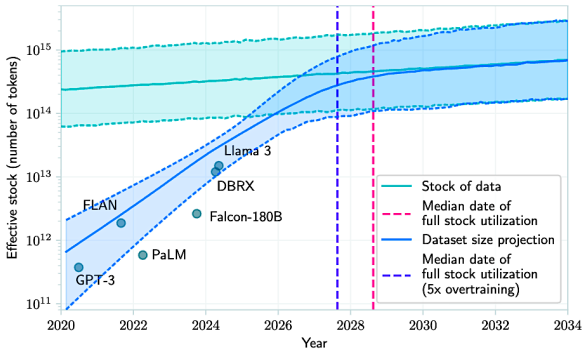


Figure 1: **Forecasted exhaustion of public human text for LLM training.** Teal: estimated effective stock of quality-adjusted, de-duplicated public human text, ≈ 4–5×10^{14} tokens. Blue: projected dataset sizes of frontier runs. Vertical markers: median crossover dates. **Takeaway:** demand meets supply around the late 2020s, motivating controlled synthetic-data regimes. Reproduced from Villalobos et al. (2024).

Synthetic recursion is increasingly unavoidable. Synthetic corpora already fill domain gaps and augment instruction tuning, while future crawls increasingly contain model-generated text, creating ***recursive training loops*** whether or not anyone intends them. Synthetic recursion is ***not only a design choice, but an emerging property of the data ecosystem***.

Collapse is a distributional failure. Naïve recursive training induces *model collapse*: rare modes disappear, diversity contracts, repetition rises and errors compound (Shumailov et al., 2024; Alemohammad et al., 2024; Dohmatob et al., 2024; Guo et al., 2024). These failures appear first in the *tails* and, when a model labels its own data, in the *labels*. Collapse is thus not only semantic; it involves **(i) label contamination, (ii) entropy, (iii) support coverage, (iv) novelty and (v) mechanism diversity**.

Why provenance is the wrong control knob. In mixed ecosystems, "AI vs. human" labels are noisy, adversarially gameable, and often orthogonal to information quality. A synthetic paragraph may contain useful novelty; a human paragraph may be duplicated boilerplate. The relevant question is whether the training distribution preserves the ***statistical conditions under which learning remains correct, diverse and information-rich*** (Bender et al., 2021).

Prior remedies are useful but fragmented. Tab. 1 organizes prior work into four directions.

Table 1: **Prior remedies grouped into four directions.** Each addresses part of the recursion problem, largely in isolation.

Hygiene & Mixture Quality	*Clean & decontaminate.* Deduplication, decontamination, reweighting, quality scoring. *Anchors:* C4 (Raffel et al., 2020), The Pile (Gao et al., 2020), RefinedWeb (Penedo et al., 2023), Dolma (Soldaini et al., 2024), FineWeb (Penedo et al., 2024), RedPajama (Weber et al., 2024), dedup (Lee et al., 2022; Kandpal et al., 2022), contamination (Deng et al., 2024).
Targeted Acquisition Loops	*Mine what is missing.* Uncertainty sampling, curricula, self-paced learning, prompt evolution. *Anchors:* active learning (Settles, 2009), curriculum (Bengio et al., 2009), self-paced (Kumar et al., 2010), Self-Instruct (Wang et al., 2023b), WizardLM (Xu et al., 2024), ANLI (Nie et al., 2020).
Anti-Collapse Objectives & Decoding	*Fight mode collapse.* Unlikelihood, diversity-oriented decoding, repetition penalties. *Anchors:* nucleus sampling (Holtzman et al., 2020), unlikelihood (Welleck et al., 2020), typical decoding (Meister et al., 2023), diverse beam search (Vijayakumar et al., 2018), contrastive search (Su et al., 2022).
Theory, Metrics & Verification	*Define, detect, correct.* Collapse theory, Zipf/Heaps statistics, accumulation, verifiers, correction. *Anchors:* recursive collapse (Shumailov et al., 2024; Dohmatob et al., 2024), accumulate (Gerstgrasser et al., 2025), verify (Feng et al., 2025), self-correct (Gillman et al., 2024), Zipf/Heaps (Zipf, 1949; Heaps, 1978).

The strongest mitigations keep real data in the loop (Gerstgrasser et al., 2025; Bertrand et al., 2024) or verify samples with ground truth or an oracle (Feng et al., 2025; Gillman et al., 2024); *neither helps when the clean pool is fixed and gold labels are unavailable*. What is missing is a unified, label-free protocol that jointly controls **(i) which synthetic samples enter training, (ii) whether their labels, tails and novelty stay healthy, and (iii) when the pipeline should adapt** before recursive drift compounds.

This paper. We take the view that **collapse is a property of the admitted training distribution, not of its provenance**, and operationalize it as ACP (Fig. 2). We test it in two regimes. **Regime I** is a true self-consuming loop: four students generate their own data for GSM8K, ARC-Challenge, CommonsenseQA and BoolQ, are fine-tuned on what ACP admits, and repeat for eight generations without ever seeing a gold label. **Regime II** fine-tunes four backbones on open-web text from a two-teacher lattice and audits distributional health. Our contributions are **(1)** a formal, label-free admission protocol with a consensus gate, four interlocking gates and tripwires (§ 3); **(2)** stability without gold labels—accuracy within noise in 14/16 settings while admitted precision exceeds generated precision in 128/128 rounds (§ 4.1); **(3)** evidence that accuracy is a lagging indicator, as an unfiltered loop matches ACP while its pool is 48% misla-

Approach	no gold labels	no oracle /verifier	fixed clean pool	multi-gen.	diversity gates
Accumulate (Gerstgrasser et al., 2025)	✓	✓	–	✓	–
Verify (Feng et al., 2025)	✓	–	✓	✓	–
Self-correct (Gillman et al., 2024)	✓	–	✓	✓	–
STaR (Zelikman et al., 2022)	–	✓	✓	✓	–
ReST (Gulcehre et al., 2023)	✓	–	✓	✓	–
LMSI (Huang et al., 2023)	✓	✓	✓	–	–
ACP (ours)	✓	✓	✓	✓	✓

Table 2: **Positioning.** ACP runs many generations without gold labels, an external verifier, or a growing clean pool, while explicitly gating diversity.

belled (§ 4.2); **(4)** a consensus-precision theory showing that route diversity purifies reasoning, not knowledge (§§ 3.1 and 4.3); and **(5)** distributional health on open web text, with an audit that removes an artifact from our own metric (§ 4.4).

Scope. We do not claim to solve frontier-scale synthetic pretraining; we study a controlled fine-tuning regime where collapse can be measured directly. Our claim is narrower: ***recursive fine-tuning becomes more stable under explicit, label-free admission constraints***.

2 Related Work

Model collapse. Recursive training on generated data loses tails (Shumailov et al., 2024); self-consuming loops "go MAD" without fresh data (Alemohammad et al., 2024); collapse changes scaling laws (Dohmatob et al., 2024, 2025), erodes linguistic diversity (Guo et al., 2024; Briesch et al., 2023), and can be unavoidable in some regimes (Seddik et al., 2024).

Mitigation and self-training. STaR and ReST filter rationales by correctness or reward (Zelikman et al., 2022; Gulcehre et al., 2023); LMSI fine-tunes on majority-voted answers (Huang et al., 2023; Wang et al., 2023a). These target gains in a few iterations; we study *stability over many generations* and show consensus alone is not enough (Tab. 2).

3 The Anti-Collapse Protocol (ACP)

Overview. ACP treats recursive synthetic fine-tuning as a ***closed-loop data-admission and distribution-control problem***. The key design question is not whether a sample is human or synthetic, but whether it preserves the conditions under which recursive learning remains stable: **correctness**, **learnability**, **tail support**, **coverage** and **mechanism diversity**. Let π_0 be the frozen base model and $\mathcal{S}_w$ the student after generation w. A generator (the student itself in Regime I, a teacher lattice in Regime II) answers each prompt q along K *routes* $r \in \mathcal{R}$, producing candidates $\mathcal{C}_w$. ACP performs constrained admission:

$$\mathcal{A}_w^\star = \arg\max_{\mathcal{A}\subseteq\mathcal{C}_w} |\mathcal{A}| \quad \text{s.t.} \quad \mathbf{g}(\mathcal{A} \mid \mathcal{P}_{w-1}) \succeq \mathbf{0}, \tag{1}$$

where $\mathcal{P}_{w-1} = \bigcup_{v<w} \mathcal{A}_v$ is the cumulative admitted pool and $\mathbf{g} = [g_{\text{con}}, g_{\text{ent}}, g_{\text{nov}}, g_{\text{iso}}, g_{\text{route}}]$ stacks five gates controlling, respectively, *correctness*, *learnability*, *support*, *coverage* and *mechanisms*. The student then updates recursively, $\mathcal{S}_w = \text{Train}(\mathcal{S}_{w-1}, \mathcal{P}_{w-1} \cup \mathcal{A}_w^\star)$. ACP therefore acts as a ***recursive stabilizer***, regulating the geometry of admitted information before collapse signatures compound. Gates are applied greedily in the order of Fig. 2; stateful gates are committed only for finally admitted samples. **No gate ever reads a gold label or a provenance tag.**

Consensus gate: label-free correctness. Let $a(x)$ be the extracted answer of candidate x, $\hat{a}_q$ the unique modal answer over routes, and s_q its route support. Then

$$g_{\text{con}}(x) = \mathbb{1}\big[a(x)=\hat{a}_q\big]\,\mathbb{1}\big[s_q \geq m\big]\,\mathbb{1}\big[\hat{a}_q \text{ unique}\big]. \tag{2}$$

For open-ended text, candidates are clustered by embedding cosine $\geq \tau_c$ and a cluster is admitted if it spans $\geq m$ routes.

T1: Entropy Banding. Recursive pipelines overproduce two pathological regimes: **(i)** low-entropy rote text that reinforces dominant templates, and **(ii)** high-entropy noise that injects unstable gradients. With per-token surprisal under the *frozen* base, $H_{\pi_0}(x) = -\frac{1}{|x|}\sum_i \log_2 \pi_0(t_i \mid q, t_{<i})$, T1 admits x iff $H_{\pi_0}(x) \in \mathcal{B}_w = [Q_\alpha(H_{\pi_0}), Q_{1-\alpha}(H_{\pi_0})]$ over the current candidates, acting as a ***band-pass filter on useful uncertainty***:

$$\underbrace{\text{rote repetition}}_{\text{low entropy}} \longrightarrow \underbrace{\text{learnable uncertainty}}_{\text{mid entropy}} \longrightarrow \underbrace{\text{unstable noise}}_{\text{high entropy}}$$

The middle band is, in *minimum description length* terms, compressible enough to learn yet not already known (Rissanen, 1978; Settles, 2009; Bengio et al., 2009). Anchoring to π_0 rather than $\mathcal{S}_{w-1}$ also makes the band centre a free *drift meter* (§ 4.3).

T2: Tail-Mass & Novelty Floors. Collapse appears first in the tails. With $\text{Nov}_3(x)$ the fraction of unseen 3-grams, $\text{Distinct}(x)$ the distinct-token ratio, $\text{TailMass}(x)$ the share of tokens in the rarest decile of seed types (Zipf, 1949), and $\mathcal{M}$ a memory

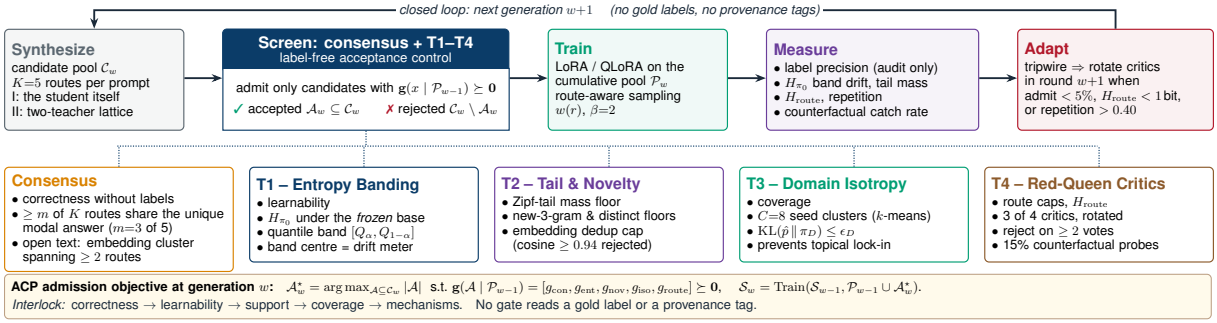


Figure 2: **ACP as closed-loop recursive control.** At generation w, candidates from several reasoning routes are screened by a label-free consensus gate and four information-geometric gates, the student is fine-tuned on the cumulative admitted pool, collapse statistics are measured, and tripwires adapt the critics before the next generation. Consensus controls **correctness**, T1 **learnability**, T2 **tail/novelty support**, T3 **coverage**, and T4 **mechanism diversity**. ACP regulates the geometry of admitted information rather than its provenance.

of admitted embeddings,

$$g_{\text{nov}}(x) = \min\big\{\mathrm{Nov}_3(x) - \gamma,\ \mathrm{Distinct}(x) - \delta,\ \mathrm{TailMass}(x) - \eta,\ \kappa - \max_{y \in \mathcal{M}} \cos(e_x, e_y)\big\}. \tag{3}$$

Tail floors keep rare support in the gradient budget, novelty floors force support expansion, and the duplicate cap κ stops paraphrastic clones (Lee et al., 2022). Together, T2 maintains ***support heterozygosity***—by analogy to population genetics, the preservation of many variants rather than clonal dominance (Nei, 1987; Hartl and Clark, 1997).

T3: Domain Isotropy. Even learnable, novel samples can concentrate on a few topics. With k-means on seed prompts defining C clusters with prior $\boldsymbol{\pi}_D$ and $\hat{\mathbf{p}}$ the pool histogram after admitting x, T3 requires $\mathrm{KL}(\hat{\mathbf{p}} \,\|\, \boldsymbol{\pi}_D) \leq \epsilon_D$, preventing style monoculture and dataset shift (Quiñonero-Candela et al., 2009; Gama et al., 2014).

T4: Red-Queen Critics over a route lattice. Collapse affects *mechanisms*, not only outputs: a model can stay correct while converging onto one scaffold. T4 caps any route's share of a round at $\rho_{\max}$ (and of the pool at $\rho_{\max}$+0.15), tracks $H_{\text{route}}(\mathcal{A}) = -\sum_r p_{\mathcal{A}}(r) \log_2 p_{\mathcal{A}}(r)$, and up-weights rare routes during training by $w(r) = 1 + (\beta - 1)(1 - n_r / \max_{r'} n_{r'})$. A panel of critics (template, sentence-length uniformity, entropy shape, lexical diversity) votes against sameness; a rotating subset is active, so no fixed style can be optimized against them—a "Red Queen" dynamic (Van Valen, 1973). A fraction of candidates is perturbed into *counterfactual probes* (shuffles, negation flips) that are scored but never trained on; their catch rate audits gate sensitivity.

Tripwires, adaptation and gate interlock. ACP monitors $\mathbf{m}(w) = [\text{admit rate}, H_{\text{route}}, \text{repetition}]_w$ and forces a critic rotation in round $w{+}1$ when admit rate $< 5\%$, $H_{\text{route}} < 1$ bit, or admitted bigram repetition > 0.40. The gates are intentionally interdependent:

Consensus: correct → **T1**: learnable → **T2**: support → **T3**: coverage → **T4**: mechanisms.

Consensus purifies labels; T1 keeps them learnable; T2 preserves rare support; T3 spreads it across topics; T4 keeps plural routes—which, as we show next, is exactly what consensus needs to work. Gate-by-gate rationale and real-data diagnostics are in § A.

3.1 Why consensus works, and when it does not

Let K routes answer correctly with probability p each and spread wrong answers uniformly over L alternatives (L=1 yes/no, L=3 four-way choice, $L{\to}\infty$ free-form numbers).

Proposition 1 (Consensus precision)**.** *If route answers are conditionally independent given the prompt, the precision of m-of-K consensus satisfies*

$$\pi_\perp(p) \ \geq\ \frac{A(p)}{A(p) + L\,\bar{F}_K\big(m; \frac{1-p}{L}\big)}, \tag{4}$$

where $A(p) = \bar{F}_K(m; p)$ and $\bar{F}_K(m; q) = \Pr[\mathrm{Bin}(K, q) \geq m]$. As $L \to \infty$, $\pi_\perp \to 1$, so the lift $\pi_\perp - p \to 1 - p$ grows as the generator weakens.

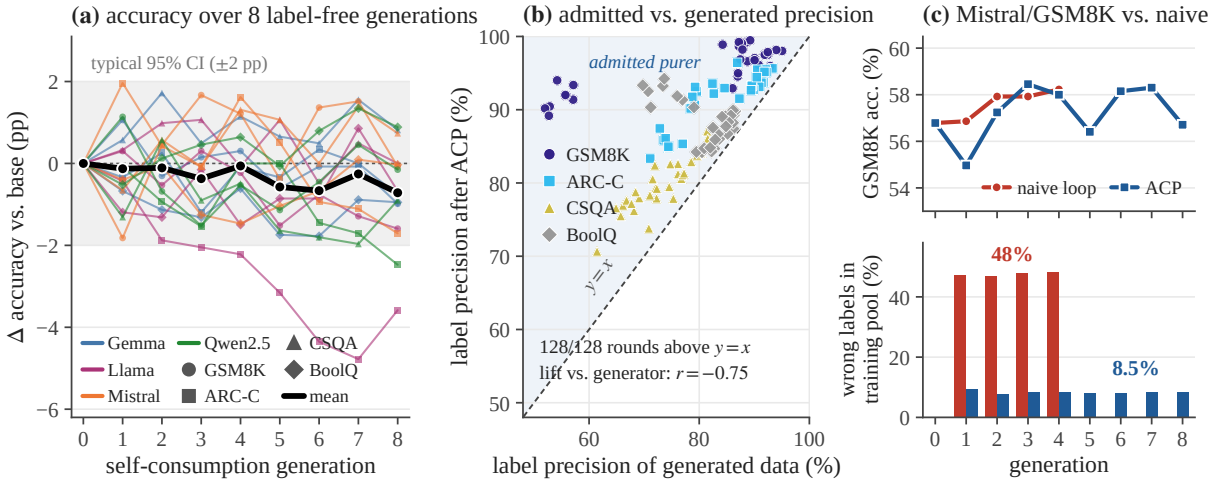


Figure 3: **Eight generations of label-free self-consumption. (a)** Accuracy change relative to the base model for 16 model–task settings (colour: model; marker: task); bold: mean; shaded: a typical 95% CI. **(b)** Label precision of admitted vs. generated data for all 128 rounds: every point lies above $y{=}x$, and the lift grows as generator precision falls. **(c)** Against an unfiltered loop (Mistral/GSM8K), accuracy is indistinguishable (top) while 48% of the naive pool is mislabelled versus 8.5% under ACP (bottom).

The proof is a union bound (§ B.2). Routes of one model are *not* independent: a knowledge gap affects every route. Modelling a fraction ρ of prompts as *common-mode* (all routes share one answer, correct w.p. p),

$$\pi_\rho = \frac{\rho p + (1-\rho)A(p)}{\rho + (1-\rho)\big(A(p) + B(p)\big)}, \qquad (5)$$

with $B(p)$ the wrong-consensus term of Eq. (4). Inverting Eq. (5) at the observed precision gives $\hat{\rho}$. Since $\pi_\rho \to p$ as $\rho \to 1$, ***consensus can only purify what routes get wrong independently***—precisely what T4 protects.

4 Experiments and Results

We test ACP under ***recursive synthetic reuse***. The question is simple: ***does controlled admission prevent collapse before headline accuracy fails?***

Experimental Setup. Regime I (self-consuming). *Students:* Gemma-2-9B-it (Gemma Team, 2024), Llama-3.1-8B-Instruct (Grattafiori et al., 2024), Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), Qwen2.5-7B-Instruct (Qwen Team, 2024). *Tasks:* GSM8K (Cobbe et al., 2021), ARC-C (Clark et al., 2018), CSQA (Talmor et al., 2019), BoolQ (Clark et al., 2019). Each generation the current student answers 400 fresh questions along $K{=}5$ routes (2,000 candidates), ACP filters with $m{=}3$ and a 10–90th-percentile band, and the student is LoRA-tuned (Hu et al., 2021) on the cumulative pool; 8 generations, greedy evaluation on full splits. **Gold labels are used only for evaluation and the diagnostic "label precision".** Reference: an unfiltered *naive* loop for Mistral/GSM8K.

Regime II (open web). Four base backbones, 4-bit QLoRA (Dettmers et al., 2023); 40 FineWeb seeds per round (Penedo et al., 2024); two teachers (Qwen2.5-7B-I, Mistral-7B-I) × five routes = 400 candidates; MiniLM consensus ($\tau_c{=}0.55$, $m{=}2$) (Wang et al., 2020; Reimers and Gurevych, 2019); 20–80th-percentile band. Seeded teachers mean *all four students screen the same pools*. Full settings: § B.1.

4.1 Main results: eight generations without gold labels

Figs. 3 and 4 and Tab. 3 summarize the central result: ***recursive self-training does not inevitably collapse if data admission is treated as a closed-loop control problem***. It is best read in three parts. **(i) Collapse appears before headline accuracy fails.** Self-consumption *does* degrade the student, most clearly in the precision of its own outputs: generated-label precision trends downward over rounds 1–8 in 14 of 16 settings (mean linear trend −2.6 pp, up to −7.7 pp for Llama/BoolQ; Fig. 4, SYNTHESIZE). Accuracy barely moves: the mean change after eight generations is −0.72 pp (mean $|\Delta|$ 1.02 pp), and 14/16 settings show no significant change. The two exceptions, Llama/ARC-C (−3.6 pp, $p{=}0.014$) and Qwen/ARC-C (−2.5 pp, $p{=}0.044$), do not survive correction. Outside ARC-

		Accuracy (%)					Admitted data			
Student	**Task**	G0	G4	G8	Δ_8	p	gen. prec.	adm. prec.	noise ↓	admit
Gemma-2-9B	GSM8K	88.9	89.2	87.9	−1.0	.43	88.9	95.8	62%	26%
	ARC-C	91.7	91.6	91.0	−0.8	.51	91.7	94.5	34%	59%
	CSQA	79.0	80.1	79.8	+0.8	.62	75.7	80.1	18%	71%
	BoolQ	88.1	87.5	87.2	−0.9	.24	85.9	89.1	22%	69%
Llama-3.1-8B	GSM8K	85.9	85.7	84.3	−1.6	.25	87.4	98.9	91%	53%
	ARC-C	87.2	85.0	83.6	−3.6	.01	81.7	92.9	61%	61%
	CSQA	77.4	77.3	77.4	+0.0	1.0	72.0	80.2	29%	66%
	BoolQ	81.9	80.4	81.2	−0.7	.48	73.8	92.2	70%	48%
Mistral-7B	GSM8K	56.8	58.0	56.7	−0.1	.97	54.7	91.6	81%	34%
	ARC-C	77.0	78.7	75.3	−1.7	.33	74.5	86.0	45%	58%
	CSQA	70.5	71.8	71.3	+0.7	.69	66.5	76.3	29%	63%
	BoolQ	83.5	82.0	83.5	−0.0	.97	82.6	86.1	20%	61%
Qwen2.5-7B	GSM8K	90.3	89.8	90.1	−0.2	.90	92.3	97.4	67%	59%
	ARC-C	91.5	91.5	89.0	−2.5	.04	89.7	93.3	35%	66%
	CSQA	82.2	81.7	81.3	−0.9	.56	80.2	84.2	20%	71%
	BoolQ	84.3	85.0	85.2	+0.9	.32	83.2	86.3	18%	68%
Mean		82.3	82.2	81.6	**−0.72**		80.0	**89.0**	**44%**	58%

Table 3: **Regime I: accuracy and admitted-data quality after 0, 4 and 8 self-consuming generations.** p: two-sided two-proportion test of G8 vs. G0 (conservative, unpaired); none is significant after Bonferroni correction (α=0.05/16). Precisions are averaged over the 8 rounds; "noise ↓" is the relative reduction of mislabelled samples, $1-(1-\text{adm.})/(1-\text{gen.})$; "admit" is the share of the 2,000 candidates admitted per round.

C the mean is −0.24 pp, well inside the evaluation sets' 95% half-widths (±1.1–±2.7 pp).

(ii) ACP acts on the cascade, not one metric. Admitted-label precision exceeds generated precision in **128 of 128** rounds (Fig. 3b), raising macro precision from 80.0% to 89.0%—a cut in label noise from 20.0% to 11.0%. Admitted precision stays nearly flat (mean trend −0.4 pp), so ACP *absorbs about 83% of the generator's erosion* before it reaches the training pool. The lift is strongly anti-correlated with generator quality (round-level $r=-0.75$, $p<10^{-24}$), as Proposition 1 predicts. Mistral/GSM8K, whose own samples are 55% correct, trains on data that is 92% correct; Llama/BoolQ's outputs fall from 79% to 74% correct while its training data rises from 90% to 94%.

(iii) Stability does not come from collapsing diversity. Admitted routes stay near-uniform throughout (route entropy 2.30 of a maximal 2.32 bits), counterfactual probes are caught at 0.54–0.73, and no tripwire fires in Regime I (Fig. 4). ACP thus preserves *multiple reasoning mechanisms*, not one surface-level scaffold.

ARC-C is the one consistent weak spot. All four students finish below base on ARC-C (mean −2.1 pp vs. −0.24 pp elsewhere; Welch p=0.04). It is the only task whose prompt pool (1,418) is smaller than the 3,200 prompts consumed, so from round 4 the loop re-answers questions it has trained on—a miniature of data exhaustion—although the decline starts earlier (−1.2 pp by G3).

What ACP certifies (and what it does not). Labels. Admitted > generated precision in 128/128 rounds; noise −45% relative. **Learnability.** Every admitted sample lies in a H_{π_0} band anchored to π_0. **Support.** Regime II tail mass ↑ in 31/32 backbone-rounds. **Mechanisms.** Route entropy ≥97% of maximum; no tripwire in Regime I. **Not certified:** T3 and critic votes never rejected a sample at our settings; their role here was monitoring, not filtering (§ 4.5).

4.2 Accuracy is a lagging indicator of collapse

Same accuracy, different data. The naive loop admits everything and, after four generations, scores 58.2% on GSM8K versus 58.0% for ACP (Fig. 3c). Judged by accuracy alone, filtering looks unnecessary. The training pools tell a different story: by generation 4 the naive pool holds 3,849 mislabelled examples (48.1%) against 233 (8.5%) for ACP—a 16.5× difference—while ACP trains on only 34% as many samples. Networks fit wrong labels before test accuracy reveals it (Zhang et al., 2017); the naive pool's higher loss (0.161 vs. 0.123) shows the contamination being absorbed.

Accuracy vs. Collapse Telemetry. Accuracy. Naive 58.2% vs. ACP 58.0% at

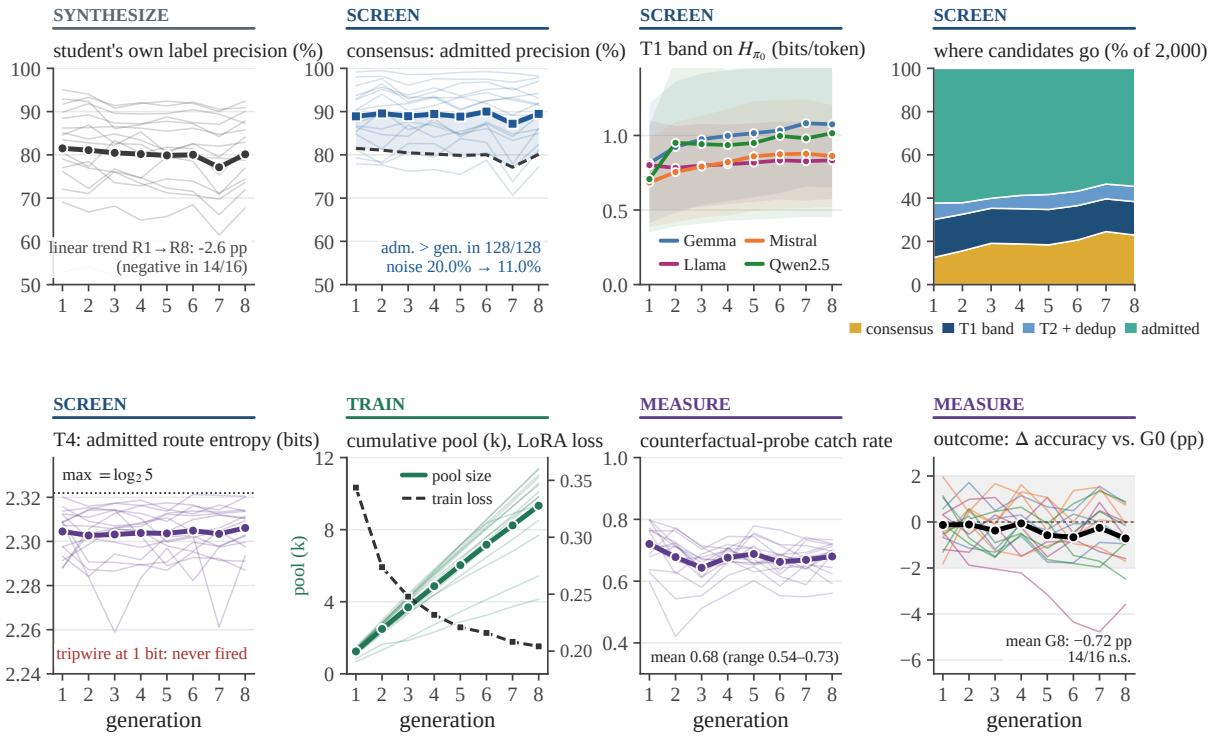


Figure 4: **ACP recursive dynamics dashboard (Regime I, 16 settings × 8 generations).** Tags mark the loop stage; thin lines are settings, bold lines their mean. SYNTHESIZE: the student's own label precision erodes. SCREEN: consensus keeps admitted precision flat and above generated (shaded gap); the T1 band on H_{π_0} drifts upward for every student; consensus and T1 do most of the rejection; T4 keeps route entropy within 3% of its maximum. TRAIN: the cumulative pool grows linearly while loss falls. MEASURE: counterfactual probes are caught at a steady rate, and accuracy stays near its base.

G4—indistinguishable (±2.7 pp CI). **Telemetry.** Naive pool: 48.1% wrong labels; ACP pool: 8.5%. ***A self-consuming loop can look healthy on its benchmark while its training distribution is already half wrong***, so collapse must be monitored—and prevented—at admission, not at evaluation.

4.3 What consensus can and cannot fix

Routes decorrelate reasoning, not knowledge. Under independence, Eq. (4) predicts admitted precision of 0.96–1.00 in 15/16 settings; we observe 0.76–0.99. Inverting Eq. (5) yields the same ordering for every student (Fig. 5a): GSM8K ($\hat{\rho}$=0.09–0.38) < ARC-C (0.36–0.66) < CSQA (0.64–0.80), with BoolQ high (0.73–0.76) for three students. For Llama/BoolQ $\hat{\rho}$=−0.26: because precision is logged after *all* gates, T1 and deduplication remove errors beyond consensus. Arithmetic errors depend on the reasoning path, so another route often avoids them; knowledge errors are shared by every route. Hence purification is strongest on GSM8K (62–91% noise reduction) and weakest on CSQA (18–29%); the lever for further gains is ***decorrelating routes***—the role of the T4 lattice.

Drift is not collapse. Because T1 is anchored to π_0, the band centre measures how surprising the student's text is to its base. It rises in 15/16 settings, by up to 84% (Fig. 5c), yet drift is unrelated to accuracy change (r=0.25, p=0.36). ACP permits distributional movement while blocking the failure modes—contamination, duplication, route collapse—that turn drift into degradation.

4.4 Distributional health on open web text

Tail support is enriched. Admitted text carries 1.13–1.31× the rare-token mass of the generated pool in 31/32 backbone-rounds (Fig. 6; per-backbone summary in Tab. 9, § B.5). Tail mass is a per-token rate, so pool size does not confound it (Shumailov et al., 2024).

The lattice stays plural. Per-cell admission varies from 21% to 63%, so the gates are selective about mechanism; yet the admitted pool retains 97.6–98.5% of maximal teacher–route entropy, and teacher identity carries ≤0.007 bits about route.

Tripwires adapt where needed. Mistral-7B's repetition tripwire fired in all eight rounds; the

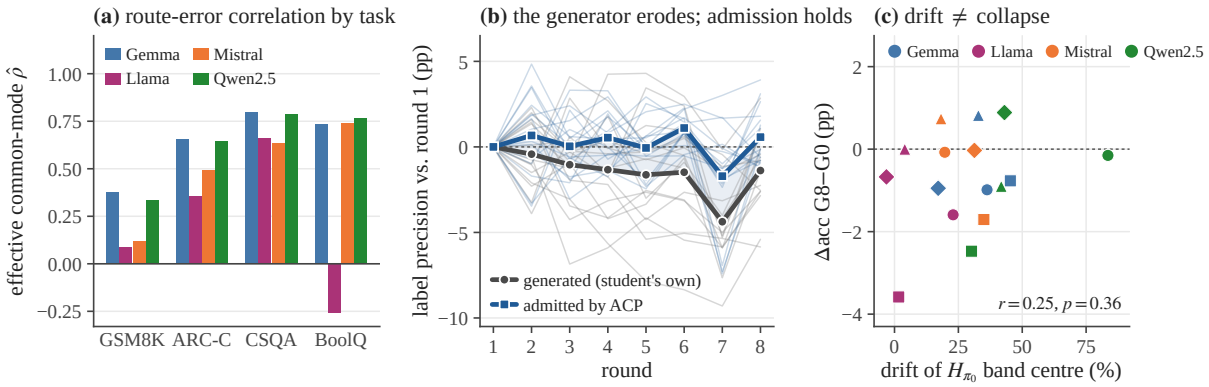


Figure 5: **Mechanisms. (a)** Effective common-mode error fraction $\hat{\rho}$ (Eq. (5)): route errors are least correlated on multi-step math and most correlated on knowledge-heavy tasks, for every student. **(b)** Label precision relative to round 1 (thin: settings; bold: mean): the student's own outputs erode, admitted data does not. **(c)** The T1 band drifts by -3 to $+84\%$ under self-consumption, yet drift is unrelated to accuracy change.

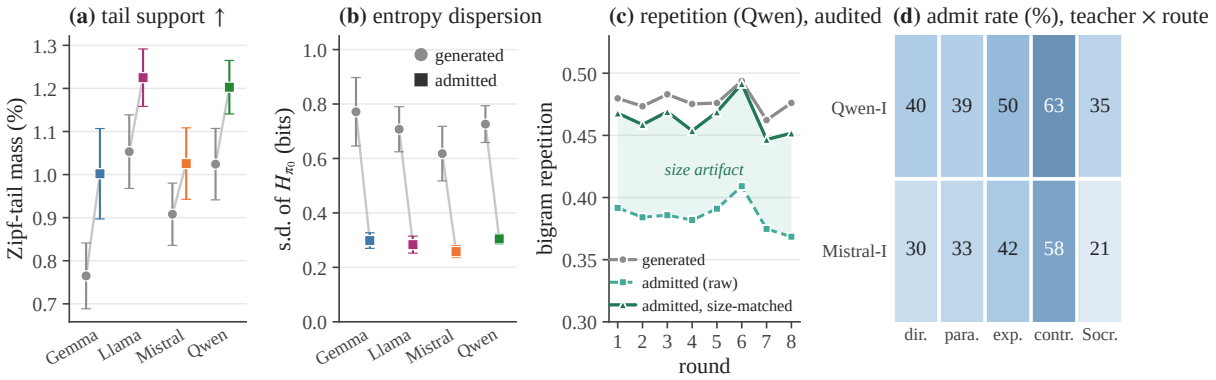


Figure 6: **Regime II: generated vs. admitted data, four backbones × 8 rounds** (mean ± s.d. over rounds). **(a)** Admitted text carries 13–31% more Zipf-tail mass. **(b)** T1 compresses entropy dispersion (by construction; a sanity check). **(c)** Raw repetition suggests a large gain, but after size-matching most of it disappears; a small residual remains in 8/8 rounds. **(d)** Admission rates differ up to 3× across teacher×route cells, yet the admitted lattice retains ≥97.6% of its maximal joint entropy.

other backbones tripped once each, all in round 6, as expected from their shared candidate pools.

Auditing our own metric. Admitted repetition looks 0.09 lower than generated, but admitted pools are about half the size and bigram repetition grows with corpus size. Size-matching (§ B.5) shows that **85% of the apparent gain is an artifact**; a genuine reduction of 0.014 remains, positive in 8/8 rounds. Likewise, the entropy-dispersion reduction (Fig. 6b) follows from T1's design and is not a finding.

4.5 Which gates bind?

Consensus and T1 do most of the filtering (Fig. 14 in § B.4): in Regime I, consensus removes 19% of candidates, T1 16%, dedup 5% and T2 1%; in Regime II, T1 removes 35% and T2 11%, because open text varies far more in novelty. The T3 cap and the critic vote threshold *never bound* at our scale; their contribution here is monitoring (counterfactual catch rates 0.42–0.80). The stability we observe is therefore attributable chiefly to consensus, T1, T2 and deduplication.

5 Discussion and Conclusion

Synthetic repetition is not the enemy; uncontrolled repetition is. What changes under self-consumption are properties of the admitted data, none of which requires knowing who wrote a sample. We introduced ACP, a label-free, provenance-agnostic protocol that treats recursive fine-tuning as constrained data admission. It holds accuracy within noise in 14/16 settings and purifies its supervision in every round.

Monitor the training pool, not only the benchmark: collapse is an ***auditable control problem***, controlled where it begins—in what a model is allowed to learn from.

Limitations

ACP should be read as a ***controlled anti-collapse protocol*** for recursive synthetic fine-tuning, not as a universal proof that synthetic recursion is safe. Recursive replacement can induce collapse (Shumailov et al., 2024; Alemohammad et al., 2024; Dohmatob et al., 2024), while accumulation, verification and correction can mitigate it under specific assumptions (Gerstgrasser et al., 2025; Gillman et al., 2024; Feng et al., 2025). ACP does not deny collapse risk; it makes the risk ***auditable, gated and intervention-triggered***.

i) Controlled recursion, not frontier-scale pre-training. We study 7–9B students with LoRA/QLoRA adaptation and a fixed protocol. Collapse may emerge differently at larger scale, in trillion-token pre-training, or in continuously refreshed web crawls.

ii) Finite recursive horizon. We run eight generations of 2,000 (Regime I) or 400 (Regime II) candidates. Longer chains may expose delayed tail erosion or creeping drift; we do not claim that ACP sustains indefinite self-training.

iii) Diagnostics are partly protocol-internal. Entropy-dispersion compression follows from T1 by design, and most of the raw repetition gain is a size artifact (§ B.5). Label precision is external—it uses held-out gold labels never seen by the gates—but Regime II evaluates data health, not downstream task accuracy.

iv) Threshold sensitivity. ACP depends on explicit thresholds (m, α, γ, δ, η, κ, ϵ_D, $\rho_{\max}$). We did not run threshold sweeps; learning adaptive thresholds or deriving stability conditions for $\mathbf{g}(\mathcal{A}) \succeq \mathbf{0}$ remains open.

v) Baselines and causal attribution. The unfiltered loop was run for one model–task pair and four generations. A full naive and per-gate ablation grid is needed to attribute stability to individual gates.

vi) Correlated errors bound consensus. Consensus cannot remove errors shared across routes ($\hat{\rho}$ up to 0.80 on knowledge tasks). The common-mode model of Eq. (5) is a one-parameter summary, and $\hat{\rho}$ uses per-sample precision as a proxy for per-prompt behaviour.

vii) Gate coverage. T3 and the critic votes did not bind at our settings, so we cannot yet quantify their filtering value. Temporal stratification and teacher-divergence terms discussed in § A were not exercised.

viii) Prompt reuse. ARC-C's prompt pool is exhausted after three rounds, and it is the only task with a consistent decline. A fresh-prompt requirement or recycling-aware gate should precede longer runs.

ix) Statistics. Per-item predictions were not logged for every evaluation, so significance tests are unpaired and conservative.

x) Collapse scope is deliberately bounded. ACP targets distributional and label collapse. It does not address factual contamination beyond consensus, privacy leakage, bias amplification, adversarial poisoning or reward hacking.

Ethics Statement

All data are public benchmarks or public web text (FineWeb). Self-generated training data can propagate model errors and biases; ACP reduces label noise but does not certify factual correctness, and models trained this way should be evaluated before deployment.

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Frequently Asked Questions (FAQs)

✻ What is the strongest bounded claim of ACP?

➠ ACP does not claim that synthetic recursion is universally safe, nor that model collapse is solved at frontier scale. The central claim is narrower and more auditable: Recursive synthetic fine-tuning need not collapse when admitted data are controlled for correctness, learnability, tail support, coverage and mechanism diversity—*without gold labels*. In our controlled setting, ACP keeps accuracy within noise in 14/16 settings for eight generations and purifies its own supervision in 128/128 rounds. This is deliberately weaker than "synthetic data are safe," but stronger than a diagnostic-only study: ACP is an intervention protocol with gates, tripwires and adaptation.

✻ Why is ACP not just another data filter?

➠ ACP is a recursive closed-loop control protocol. A static filter accepts or rejects samples once; ACP repeatedly synthesizes, screens, trains, measures and adapts. This matters because collapse is *temporally compounded*: in Regime I the generator at generation w is the student trained on everything admitted before it, so each admission decision changes the next candidate pool. The question is not only which samples enter training, but how they shape the data the model will produce next.

✻ Is ACP simply self-consistency filtering?

➠ No. Consensus is one of five gates. In Regime I it removes 19% of candidates, while T1, T2 and deduplication remove another 23%. More importantly, our analysis of *when* consensus works (§ 4.3) is new: its benefit is bounded by route-error correlation $\hat{\rho}$, which is task-dependent and motivates the T4 route lattice.

✻ Does ACP merely preserve the diagnostics it was designed to preserve?

➠ We agree that preserving protocol-internal quantities alone would be insufficient, and we separate the two kinds of evidence:

- **External:** accuracy on full evaluation splits and label precision measured with held-out gold labels that no gate ever reads.
- **Partly internal:** entropy-dispersion compression follows from T1 by design, and we say so; 85% of the raw repetition gain turns out to be a pool-size artifact, and we report the corrected value (§ B.5).
- **Not targeted:** teacher–route balance in Regime II is preserved although the route caps never bound.

✻ How is ACP different from provenance filtering or AI-text detection?

➠ ACP rejects the premise that provenance is the right control variable. A synthetic paragraph can contain useful novelty; a human paragraph can be duplicated boilerplate. In mixed future crawls, "AI vs. human" labels are noisy, gameable and increasingly unavailable. ACP instead controls the geometry of admitted information:

- **Consensus:** correctness via route agreement;
- **T1:** learnability via entropy banding;
- **T2:** support via tail mass, novelty and duplicate caps;
- **T3:** coverage via cluster isotropy;
- **T4:** mechanism plurality via Red-Queen critics.

✻ The unfiltered loop did not collapse in accuracy. Does that undermine the paper?

➠ No—it is one of our main findings. Within four generations, accuracy cannot distinguish a loop training on 48% wrong labels from one training on 8.5%. That is exactly why admission-time control is needed: by the time a benchmark registers the damage, the training distribution is already contaminated. We do not claim the naive loop would collapse at a particular generation; we claim accuracy is a lagging and insensitive signal.

✻ Are eight generations enough to prove collapse prevention?

➠ No. Eight generations cannot rule out all delayed collapse modes. We interpret them as a stress test, not as proof of asymptotic stability. The informative observation is that the generator's own label precision already erodes within this horizon (14/16 settings), so the horizon is long enough to exercise the protocol. Longer runs, with a fresh-prompt requirement, are the natural next step.

✻ Are the gains simply due to a diverse teacher lattice?

➠ Not in Regime I, where there is no teacher at all: each student generates its own candidates, and diversity comes only from its five prompting routes.

Regime II does use two teachers, and there the comparison of interest is generated vs. admitted data from the *same* teachers. Stronger separation would add single-teacher and gate-only baselines.

✻ Are ACP thresholds hand-tuned? How sensitive is it to them?

➠ The thresholds are explicit and fixed in advance (Tab. 5), not tuned per setting: the same m, band quantiles, floors and caps are used for all 16 Regime I settings. We do not claim they are optimal and did not run sweeps. The most informative sensitivity plots would be: consensus m vs. admitted precision and yield; band width vs. accuracy and drift; novelty floor vs. admitted precision; duplicate cap vs. repetition and data efficiency; and ϵ_D vs. coverage. We expect a stable middle regime: too-loose thresholds recover naive-loop contamination, while too-tight thresholds starve the loop of data.

✻ Why not simply use deduplication, quality filtering or mixture rebalancing?

➠ These methods are useful but incomplete. Deduplication controls clone inflation but not label noise or route monoculture; quality filtering may overselect head templates; mixture rebalancing improves coverage but guarantees neither correctness nor tail support. ACP coordinates them in one interlock—correct → learnable → support → coverage → mechanisms—because recursive collapse is multi-axis.

✻ Why not use gold labels or a verifier?

➠ Self-improving systems will increasingly face data without labels. Methods that assume a verifier (Feng et al., 2025) or a correction oracle (Gillman et al., 2024) are complementary and could replace or augment our consensus gate when available.

✻ Do T3 and the critics matter if they never bound?

➠ At our scale and horizon they acted as monitors, not filters. We report this rather than claim an effect we did not measure. Quantifying their filtering value requires longer horizons, narrower generators and per-gate ablations. § A shows their monitoring signals on real logs.

✻ Why does ARC-C decline for every student?

➠ ARC-C combines two properties. Its common-mode error rate is moderate ($\hat{\rho}$ 0.36–0.66), so consensus removes only part of its label noise. And it is the only task whose prompt pool (1,418 questions) runs out within eight rounds, so from round 4 the loop re-answers, and re-trains on, questions it has already seen. Neither property alone explains the timing, since the decline begins at G1–G3. We see ARC-C as the earliest observable instance of the exhaustion regime ACP is designed for.

✻ How does ACP relate to work claiming collapse is inevitable or avoidable?

➠ ACP sits between the two views. Collapse studies show that recursive replacement depletes tails and amplifies errors (Shumailov et al., 2024; Dohmatob et al., 2024); other work shows accumulation, verification or correction can mitigate it (Gerstgrasser et al., 2025; Feng et al., 2025). ACP adopts the second interpretation without assuming a clean pool or an oracle: collapse is not denied, but treated as measurable and controllable at admission.

✻ Does ACP prove that synthetic data can replace human data?

➠ No. ACP is not a replacement argument. It addresses the realistic setting where synthetic data are increasingly present in training mixtures, intentionally or through contaminated crawls. The goal is to make recursive reuse safer and auditable, not to eliminate the need for human-authored, grounded, expert data.

✻ Does preserving novelty risk admitting hallucinations?

➠ Yes. Novelty is not automatically useful. ACP mitigates this by coupling novelty with other gates: consensus requires several routes to agree; T1 rejects high-surprisal noise; T2 requires novelty without duplicate inflation; T4 discourages single-route artifacts. But consensus cannot remove errors shared by every route ($\hat{\rho}$ up to 0.80), so retrieval verification and grounding-aware novelty are natural extensions.

✻ Does route entropy really measure mechanism diversity?

➠ Route entropy is a proxy, not a ground-truth measurement of internal computation. Its purpose is to prevent recursive supervision from collapsing onto a single prompt scaffold. The consensus analysis gives an indirect, more functional signal: routes are useful exactly to the extent that their *errors* are decorrelated, which $\hat{\rho}$ measures. Representation-level probes would be a stronger test.

✻ Can ACP handle adversarial or malicious

synthetic data?

➠ Only partially. ACP can suppress some adversarial effects if they appear as entropy outliers, duplicate bursts, topical drift or route collapse, but it is not a defense against targeted poisoning, hidden triggers or privacy leakage. It should be combined with contamination audits, retrieval-grounded verification and robust training. ACP is a collapse-control layer, not a security layer.

✻ What external evaluations would most strengthen ACP?

➠ Evaluations where success cannot be reduced to ACP's own gates: rare-entity factual recall; multilingual and dialectal tail slices (our pilot, § B.6, is a first step); long-tail QA and calibration; contaminated-crawl simulations mixing human, synthetic and duplicated text; and longer recursion with fresh prompts.

✻ What would falsify ACP's central claim?

➠ ACP would be weakened if a matched naive or hygiene-only loop preserved the same admitted label precision, tail mass and route diversity without ACP gates, or if ACP maintained its diagnostics but lost accuracy over longer horizons. A clean test: run naive, dedup-only, consensus-only and full ACP loops under identical budgets for 16+ generations with fresh prompts, and compare both training-pool telemetry and held-out accuracy.

✻ What are the computational costs of ACP?

➠ Screening is cheap relative to generation and training. In Regime I, filtering 2,000 candidates (answer extraction, consensus, H_{π_0} scoring under the frozen base with adapters disabled, embeddings, dedup, critics) took 35 s per round on one H100, against 52 s of LoRA training. The dominant extra cost is generating $K{=}5$ routes per prompt. ACP spends this computation before training to avoid spending gradient budget on contaminated data—the naive loop trained on 2.9$\times$ more samples, 48% of them wrong.

✻ Why is "heterozygosity" the right metaphor?

➠ We use it as an analogy for preserving multiple viable variants rather than allowing clonal dominance (Nei, 1987; Hartl and Clark, 1997). In ACP the variants are informational modes: correct answers reached by different routes, rare tokens, unseen n-grams, topical clusters and reasoning mechanisms. A model can preserve headline performance while losing these, which is why ACP monitors them directly.

✻ How should readers interpret the tripwires?

➠ A tripwire is not a proof that collapse is impossible; it is a pre-specified, auditable trigger. ACP monitors $\mathbf{m}(w) = [\text{admit rate}, H_{\text{route}}, \text{repetition}]_w$ and rotates critics when a threshold is crossed. In Regime I no tripwire fired; in Regime II, Mistral-7B's repetition tripwire fired every round while the other backbones tripped once each, in the same round, as expected from their shared candidate pools.

✻ One-sentence summary.

➠ ACP does not claim that synthetic recursion is harmless; it shows that, in a controlled fine-tuning regime and without a single gold label, the collapse cascade can be made measurable, gated and interruptible—and that a benchmark alone would not have noticed it starting.

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A The Anti-Collapse Protocol in Depth

Section goal. We motivate a provenance-agnostic, information-geometry-driven protocol for preventing collapse during repeated synthetic fine-tuning. Rather than policing authorship ("AI vs. human"), ACP regulates the *shape* of information that enters training. It imposes acceptance constraints on correctness, entropy, tails, novelty, coverage and mechanism diversity; couples them to auditable diagnostics with tripwires; and runs as a closed loop, SYNTHESIZE → SCREEN → TRAIN → MEASURE → ADAPT (Fig. 2). This stance aligns with evidence that degeneration is caused by recursive exposure to impoverished signal, not by provenance per se (Shumailov et al., 2024; Bender et al., 2021). For each gate we separate the *design* (the general control law) from what was *actually instantiated* in our experiments, and we show the gate's behaviour on the real logs.

A.1 Provenance-agnostic control via information geometry

A minimal mathematical framing. For a sequence $x = (t_1, \ldots, t_{|x|})$ with conditionals $p(t_i \mid x_{<i})$, define the mean per-token surprisal

$$H_{\text{seq}}(x) = -\frac{1}{|x|} \sum_{i=1}^{|x|} \log p(t_i \mid x_{<i}).$$

Consider a generative model M with token distribution p_M over a vocabulary $\mathcal{V}$. In repeated synthetic fine-tuning, failure manifests as five coupled pathologies: **(i) label contamination**—when the model labels its own data, its errors become training targets; **(ii) tail death**—probability mass drifts from rare to frequent types; **(iii) support narrowing**—unique-type growth stalls despite more tokens; **(iv) style ossification**—a small set of rhetorical templates dominates; and **(v) mechanism monoculture**—many routes to an answer collapse onto one (Shumailov et al., 2024; Zipf, 1949; Heaps, 1978). Read through a common geometric lens, recursive training pushes p_M along directions that locally reduce uncertainty without introducing commensurate new support, contracting the effective information volume explored by

the model. What collapses is not only accuracy in the tails but the capacity of the data pipeline to feed gradients that *add* support rather than redistribute it toward the head.

Let the Zipf tail $\mathcal{T}_q$ contain the least frequent q-fraction of types. For any x,

$$\text{TailMass}_q(x) = \frac{1}{|x|}\sum_{t\in x} \mathbb{1}[t \in \mathcal{T}_q], \qquad \text{Novelty}_n(x) = \frac{1}{N_n(x)} \sum_{g\in \text{ngrams}_n(x)} \mathbb{1}[g \notin \mathcal{G}_n],$$

where $N_n(x)$ counts n-grams and $\mathcal{G}_n$ is the set of previously seen n-grams. In healthy corpora, vocabulary growth follows Heaps' law $V(N) \approx KN^\beta$ with $\beta \in (0, 1)$; collapse corresponds to $\beta \downarrow$ even as $N \uparrow$ (Heaps, 1978). Together these quantify how much fresh support enters training and how much useful uncertainty remains.

Common notation. Generations/rounds $w \in \{1, \ldots, 8\}$; prompt q; routes $r \in \mathcal{R}$ ($K{=}|\mathcal{R}|{=}5$); candidate pool $\mathcal{C}_w$; admitted set $\mathcal{A}_w \subseteq \mathcal{C}_w$; cumulative pool $\mathcal{P}_w = \bigcup_{v\le w} \mathcal{A}_v$; frozen base π_0; student $\mathcal{S}_w$; per-token surprisal under the base H_{π_0}; Zipf rank $r(t)$; tail set $\mathcal{T}_q$; clusters $d \in \{1, \ldots, C\}$ with prior $\boldsymbol{\pi}_D$; empirical histograms $\hat{p}$; route entropy H_{route}.

Why authorship is the wrong control knob. In modern crawls, the distinction "AI vs. human" is noisy, adversarially gameable and orthogonal to informativeness (Bender et al., 2021). A synthetic paragraph can carry high tail mass and genuine novelty; a human paragraph can be templated boilerplate. A large fraction of web text is already LLM-assisted, so provenance filtering is both brittle and incomplete. What matters is whether incoming samples (i) carry correct labels, (ii) inhabit a learnable mid-entropy band, (iii) sustain tail and novelty throughput, (iv) maintain topical coverage, and (v) preserve mechanism diversity. These are the five signals ACP regulates.

Interlocking safeguards: why the pieces work together. *Consensus* ensures that whatever enters is more likely correct than what was generated. *Entropy banding* (T1) prevents the system from drifting into a head-only regime (left tail) or accepting incoherent junk (right tail), so gradients are spent on useful uncertainty. *Tail and novelty floors with duplicate caps* (T2) convert that uncertainty into rare-token throughput while blocking inflation by paraphrastic clones. *Domain isotropy* (T3) spreads those gains across topics, stopping any single basin from hijacking the gradient field. *Red-Queen critics over a route lattice* (T4) keep *how* answers are produced plural—and, as § 3.1 shows, route plurality is the very resource consensus depends on. Together the gates implement a closed-loop control that counters the failure modes predicted by recursion theory.

Why now. Scaling theory indicates that gains at fixed compute come from informative tokens, not raw volume (Kaplan et al., 2020; Hoffmann et al., 2022). Recursion studies show that unconstrained training on generated data erodes exactly those properties (Shumailov et al., 2024). Meanwhile the web is already a mixed ecosystem, and labels for self-generated data are, by construction, unavailable. A provenance-agnostic, *label-free* control law is therefore urgent: it preserves heterozygosity while allowing synthetic pipelines to grow responsibly.

A.2 The consensus gate: label-free correctness

Intuition. When a model generates its own training data, its mistakes become training targets. Without gold labels, the only internal evidence of correctness is *agreement*: if several genuinely different reasoning routes arrive at the same answer, that answer is more likely right than any single route's output. The consensus gate turns this into an admission rule and, through Proposition 1, into a quantitative prediction of how much label noise it removes.

Rule. For each prompt, the gate computes the modal answer over the K routes, requires that it be unique and supported by at least m routes, and admits only candidates that carry it. With $m > K/2$, at most one answer can reach m votes, so uniqueness is automatic. For open-ended text, "same answer" becomes "same semantic cluster" (cosine $\ge \tau_c$ in a sentence-embedding space) and the cluster must span $\ge m$ routes.

What it can and cannot do. Proposition 1 shows that, under independent route errors, consensus precision approaches 1 as the answer space grows, and the lift over the generator grows as the generator weakens—which is exactly what we observe ($r = -0.75$ over 128 rounds). The common-mode model (Eq. (5)) shows the limit: errors shared by all routes pass through. The inferred $\hat{\rho}$ is lowest for arithmetic (0.09–0.38) and highest for knowledge-heavy tasks (0.64–0.80), so consensus is a reasoning purifier, not a knowledge verifier.

Regime I: $K=5$ routes (direct, paraphrase-first, step-by-step, contrast/verify, Socratic), $m=3$; answers parsed from `Answer:`, `\boxed{}` or the last number (GSM8K) and from the `Answer:` line (multiple choice, yes/no); truncated generations are excluded. Regime II: MiniLM embeddings, $\tau_c=0.55$, $m=2$. Consensus removes 19% of Regime I candidates on average (Fig. 15).

A.3 Technique 1: Entropy Banding — admit only learnable samples

Intuition and goal. In recursive pipelines the learner tends to waste gradient on two regimes that catalyze collapse: (i) *head rote*—spans that are trivially predictable; and (ii) *degenerate junk*—spans whose uncertainty reflects noise rather than structure. The former sharpens already-frequent templates, starving the tail; the latter injects variance without adding support. Entropy Banding institutes a provenance-agnostic control law: *admit only mid-difficulty, learnable samples*. This is the information-theoretic cousin of curriculum and self-paced learning (Bengio et al., 2009; Kumar et al., 2010) and an MDL-aligned preference for signal that is compressible but not already compressed (Rissanen, 1978).

The band and the acceptance rule. Let $\underline{h}$ and $\overline{h}$ be the α and $1-\alpha$ quantiles of H_{π_0} over the current candidates, and admit x iff $H_{\pi_0}(x) \in [\underline{h}, \overline{h}]$. Two choices matter. First, the reference is the *frozen* base π_0 rather than the evolving student: the yardstick never moves, so the band's location becomes a measurement of how far self-generated text has drifted. Second, the band is defined by *quantiles*, not fixed bits, so it adapts to languages, tokenizers and tasks with very different absolute surprisal.

Deeper intuition: what the band does to gradients. Imagine three "basins" of uncertainty. In the left basin (very low entropy) the model already assigns near-deterministic probabilities; gradients are tiny and mostly aligned with directions that amplify head templates. In the right basin (very high entropy) the model is effectively guessing; gradients are large but misaligned and dominated by noise. The middle basin contains sequences where the model has a partial hypothesis—uncertain for principled reasons. Formally, with $g(x) = \nabla \ell(\theta; x)$ and true risk gradient $G = \nabla L(\theta)$, progress per step is governed by alignment $\mathbb{E}[\langle G, g(x) \rangle]$ and variance $\mathbb{E}\|g(x)\|^2 - \|\mathbb{E}g(x)\|^2$. The left basin hurts the first term (diminishing returns); the right basin inflates the second (instability). The middle band balances them, a trade-off long noted in uncertainty sampling (Settles, 2009).

Fisher-information view. Token-level Fisher information is $I(\theta) = \mathbb{E}[\nabla \log p_\theta(y \mid x) \nabla \log p_\theta(y \mid x)^\top]$. In the left basin p_θ saturates and the score vanishes; in the right basin p_θ is near-uniform and the score is high-variance but directionally incoherent. Middle-band tokens yield peaked but non-saturated predictions whose scores have both magnitude and coherence, concentrating Fisher information along task-relevant directions. Under a fixed gradient budget, middle-band samples therefore yield the largest, most stable updates.

A toy picture. Suppose text is generated by latent "motifs" on a ring; the model starts with confidence spikes at a few angles (head modes). Left-basin updates sharpen the spikes; right-basin updates wobble aimlessly; middle-band updates—near the flanks of each spike—push the shoulders outward, widening coverage. Banding pushes probability mass *sideways* into under-covered neighbourhoods rather than upward onto covered peaks.

What we observe on the logs. Fig. 7a shows, for every Regime II backbone, the distribution of all logged admitted samples' H_{π_0} against the generated pool's logged mean $\pm$ s.d.: admitted text sits in the centre of the generated distribution with 2.4–2.6$\times$ smaller dispersion. This is a sanity check (T1 implies it), not a finding. The finding is Fig. 7b: in Regime I the band itself drifts upward for every task, i.e., the

student's own text becomes steadily more surprising to its base model—a direct, label-free measurement of self-consumption drift (up to $+84\%$), which we show is *not* predictive of accuracy loss (§ 4.3).

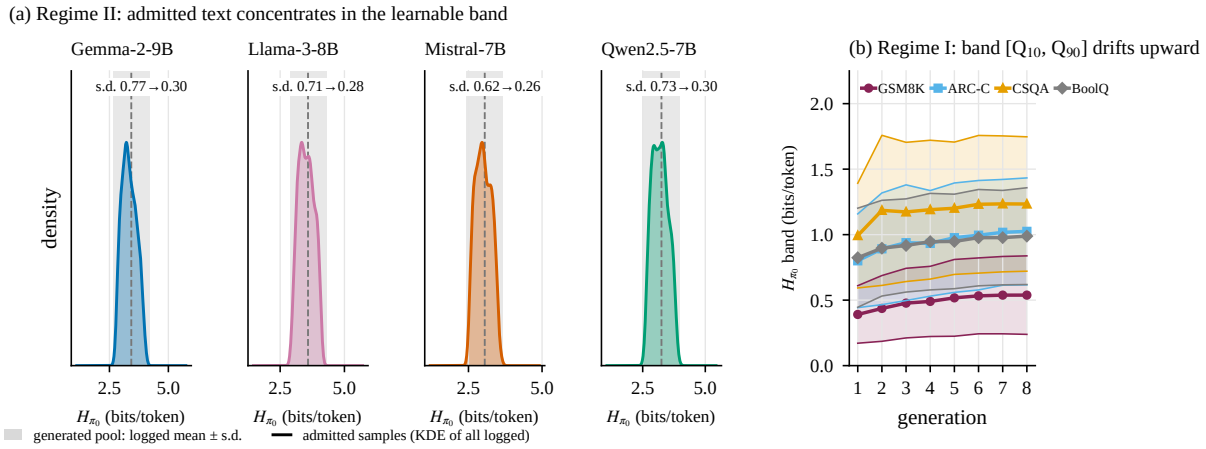


Figure 7: **T1 on real logs. (a)** Regime II: kernel density of H_{π_0} over all admitted samples (colour) against the generated pool's logged mean $\pm$ one s.d. (grey), pooled over 8 rounds; the inset gives the generated→admitted s.d. **(b)** Regime I: the band $[Q_{10}, Q_{90}]$ applied in each generation, averaged over students per task; bold lines are band centres. The upward drift is the student's text moving away from its frozen base.

Micro-failure modes to watch. *Template seepage*: sequences pass with mid entropy overall but contain low-entropy boilerplate pockets; signature: repetition rises without tail gain; fix: T2 template caps and a template critic (T4). *Noisy montage*: mid entropy arises from stitching unrelated snippets; fix: a coherence critic and a slightly lower upper edge. *Band chasing the head*: if the reference were the evolving student, quantiles would slide with it and the band could follow the head; anchoring to π_0 prevents this by construction.

Cross-lingual specifics. Entropy distributions differ by morphology and tokenization. Because the band is quantile-based, it transfers across languages; nevertheless our pilot (§ B.6) found that English-calibrated settings over-reject Devanagari text, so bands should be estimated per language or domain.

Regime I: 10th–90th percentiles of H_{π_0} over post-consensus candidates; Regime II: 20th–80th percentiles over all generated candidates. π_0 is the student with LoRA adapters disabled, so no second model copy is needed. T1 removes 16% of Regime I and 35% of Regime II candidates.

Span-level guards (sliding-window span entropy to catch hidden rote/junk pockets), a joint band over $(H_{\text{seq}}, \text{surprisal})$, and bounded centre/width nudges triggered by downstream signals are natural refinements. Reporting calibration (ECE/Brier) on admitted items would turn T1 into a calibration audit as well.

A.4 Technique 2: Tail-Mass & Novelty Floors with near-duplicate caps

Objective and rationale. Collapse shows up first in the tails: rare tokens and phrases vanish, Heaps' growth flattens, and type–token diversity gives way to paraphrastic clones. T2 establishes three coupled, provenance-agnostic controls: (i) a **Zipf-tail floor** so a fixed fraction of the update budget flows to low-frequency types; (ii) a **novelty floor** so each round contributes genuinely out-of-support n-grams; and (iii) **near-duplicate caps** that block clone inflation. Together these keep the token ecology heterozygous: rare alleles survive, novel alleles arrive, and clonal expansions are checked.

Preliminaries. Let $r(t)$ be the Zipf rank of type t in a seed sample and fix a tail quantile q (here the rarest decile). The tail set is $\mathcal{T}_q = \{t \in \mathcal{V} : r(t) > \lfloor (1-q)|\mathcal{V}| \rfloor\}$ and $\text{TailMass}_q(x)$ is the share of x's tokens in $\mathcal{T}_q$. Novelty is the share of x's 3-grams unseen in the seed sample, and the distinct-token ratio guards against degenerate repetition within a sample. Duplicates are detected in an embedding space against a sliding memory $\mathcal{M}$ of admitted samples.

Why this triad prevents tail death. Tail floors route gradient to the long tail. Novelty floors ensure the tail is replenished with unseen compositional structure rather than permutations of the head. Duplicate caps stop frequency hijacks in which semantically identical clones crowd out rare types while still passing naive novelty counters. The triad widens effective support (more types live), deepens it (more distinct n-grams), and scrambles over-represented templates.

A population-genetics picture. Think of types as alleles. Tail floors enforce a viable effective population size in the tail; novelty floors inject mutation at a controlled rate; duplicate caps limit clonal expansions (Nei, 1987; Hartl and Clark, 1997). The trio sustains heterozygosity across rounds—precisely the quantity that collapses under recursion (Shumailov et al., 2024; Dohmatob et al., 2024).

Interaction with the other gates. Consensus and T1 run first, so "novel" means *agreed-upon and learnably* out-of-support, not noise. T3 stops novelty from concentrating in one topic, and T4 makes different routes produce different n-gram frontiers for the same prompt.

What we observe on the logs. Fig. 8 plots tail mass per round for every Regime II backbone. Admitted text carries more rare-token mass than generated text in 31 of 32 backbone-rounds (1.13–1.31× on average), and the gap persists as the generated tail fluctuates. In Regime I, T2 rarely binds (1% of candidates; 0 for CSQA), because short task answers vary little in novelty, while deduplication removes 5%.

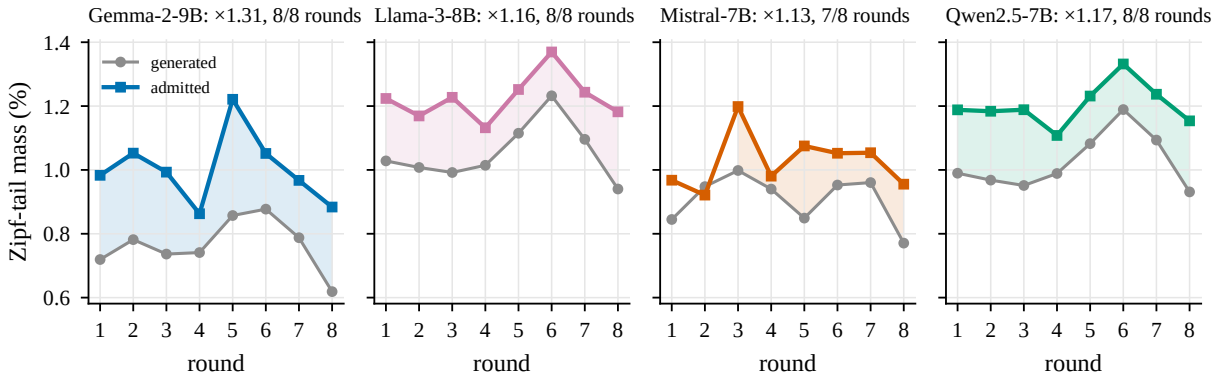


Figure 8: **T2 on real logs (Regime II).** Zipf-tail mass per round for generated (grey) and admitted (colour) text. Titles give the mean admitted/generated ratio and the number of rounds in which admitted > generated. Tail mass is a per-token rate and therefore not confounded by the smaller admitted pool.

Failure modes and mitigations. *Tail spam*: inflating tail mass with meaningless rare tokens; mitigated because consensus and T1 run first. *Template laundering*: many surface variants of one template pass novelty; mitigated by the embedding duplicate cap and the template critic. *Semantic echo*: high embedding similarity across putatively novel items; mitigated by lowering κ.

Regime I: new-3-gram rate ≥ 0.10 and distinct-token ratio ≥ 0.20; dedup at MiniLM cosine ≥ 0.94 against a 4,000-sample memory. Regime II: tail-mass floor plus new-3-gram rate $\geq 6\%$ (or tail $\geq 0.5\%$ and new $\geq 3\%$); dedup memory 2,000. T2 removes 1% (I) and 11% (II) of candidates; dedup removes 5% (I).

Separate lexical (MinHash/Jaccard) and delexicalized-template duplicate graphs with component-size caps, per-type burstiness caps, and Good–Turing estimates of unseen mass would make T2 more robust against template laundering at larger scale.

A.5 Technique 3: Domain Isotropy — preventing topical lock-in

Objective and threat model. Synthetic loops readily over-specialize: a few topics or styles dominate, the student adapts to these micro-regimes, and generalization to under-represented regions decays—classic dataset shift and concept drift (Quiñonero-Candela et al., 2009; Gama et al., 2014). T3 imposes

isotropy across topical clusters so that no single basin becomes a gravitational well. The control is provenance-agnostic: it regulates the geometry of coverage rather than who authored a sample.

Partitions and priors. Hand labels are coarse and leaky, so clusters are induced by embedding seed prompts and running k-means. Let $\boldsymbol{\pi}_D$ be the seed cluster prior and $\hat{\mathbf{p}}$ the (smoothed) cluster histogram of the pool after admitting a candidate. T3 requires $\mathrm{KL}(\hat{\mathbf{p}} \,\|\, \boldsymbol{\pi}_D) \le \epsilon_D$.

Why isotropy prevents lock-in (mechanism sketch). Under a cluster mixture $d \sim \hat{p}(d)$ the expected gradient decomposes as $\sum_d \hat{p}(d)\, G_d$ with $G_d = \mathbb{E}_{x|d} \nabla \ell(\theta; x)$. If $\hat{p}$ is skewed, learning is biased toward a few G_d directions while low-mass clusters receive insufficient corrective updates. The deviation from the target gradient is bounded by

$$\left\| \textstyle\sum_d (\hat{p}(d) - \pi_D(d)) G_d \right\| \;\le\; \|\hat{p} - \pi_D\|_1 \max_d \|G_d\| \;\le\; \sqrt{2\,\mathrm{KL}(\hat{p} \,\|\, \pi_D)}\, \max_d \|G_d\|$$

by Pinsker's inequality (Pinsker, 1964). Bounding ϵ_D therefore uniformly bounds how far updates can rotate toward a single topic.

What we observe on the logs. Fig. 9 shows the admitted coverage ratio $r(d, w) = \hat{p}(d, w)/(\hat{p}(d)\hat{p}(w))$ per cluster and round, and the per-round divergence $\mathrm{KL}(\hat{p}_w(d) \,\|\, \hat{p}(d))$ against the pooled marginal. Per-round coverage varies (each round draws only 40 seed documents over 8 clusters), but the per-round divergence stays between 0.02 and 0.14 nats—one to two orders of magnitude below ϵ_D=1.5—and the four backbones track each other because they screen the same candidate pools. This is why T3 never bound: within eight rounds the generators did not drift topically. We report it as a monitor; at longer horizons or with narrower generators it becomes a filter.

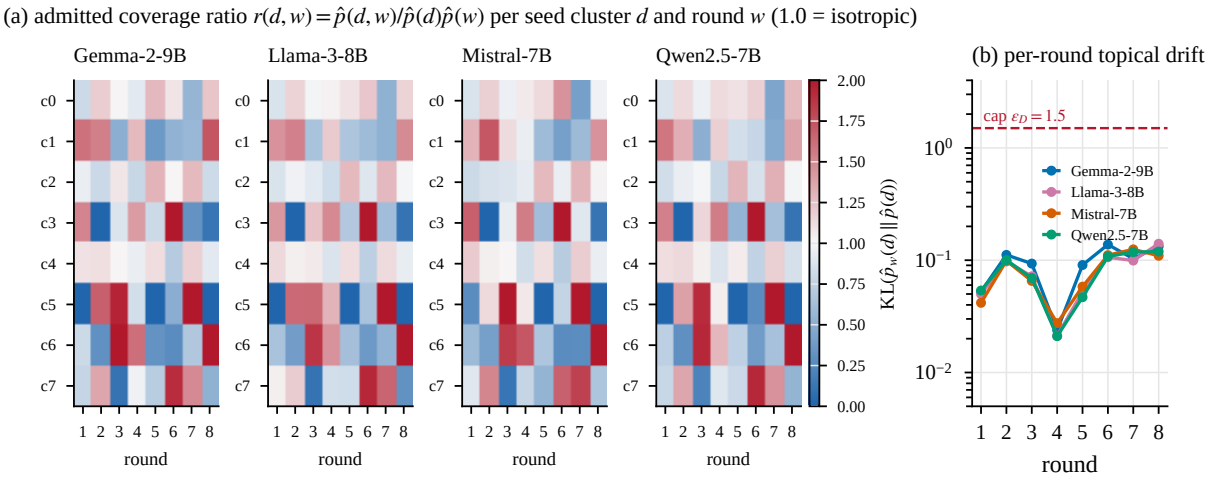


Figure 9: **T3 on real logs (Regime II). (a)** Coverage ratio of admitted samples per seed cluster d and round w relative to the pooled marginals (1.0 = isotropic). Variation reflects which 40 seed documents were drawn each round. **(b)** Per-round KL divergence of the admitted cluster distribution from the pooled distribution (log scale), far below the cap ϵ_D=1.5.

C=8 k-means clusters on MiniLM embeddings of seed prompts; ϵ_D=1.5; the same settings in both regimes. The cap never rejected a candidate.

Temporal stratification: when rounds encode time, cap $\mathrm{KL}(\hat{p}(w) \,\|\, \pi_W)$ and the joint total variation $\mathrm{TV}(\hat{p}(d, w), \pi_D \pi_W)$, or a Wasserstein-1 distance along the time axis, so that no recent synthesis wave overwhelms earlier rounds; a repair sampler can down-sample dominant cells instead of rejecting. A style-cluster entropy floor would catch "one-note" pools that look balanced by topic.

A.6 Technique 4: Red-Queen Critics over a route lattice

Objective. Recursive collapse affects not only outputs but *mechanisms*. A model can remain correct while converging onto a single rhetorical scaffold, reasoning template or teacher style. T4 preserves plural

routes through a route lattice and rotating critics, countering the diversity loss observed in recursive and repetitive generation (Shumailov et al., 2024; Holtzman et al., 2020; Vijayakumar et al., 2018).

Route lattice and route-aware training. Each prompt is answered along several routes: direct, paraphrase-first, step-by-step (expand), contrastive/verify and Socratic. T4 caps any route's share of a round and of the cumulative pool, and route-aware sampling up-weights rare routes, $w(r) = 1 + (\beta - 1)(1 - n_r / \max_{r'} n_{r'})$, so the student keeps support on multiple routes.

Critics and the Red-Queen rotation. A panel of lightweight critics votes against sameness: a *template* critic (refusal/boilerplate openings or repeated lines), a *style* critic (uniform sentence lengths), an *entropy-shape* critic (near-zero surprisal) and a *lexical-diversity* critic. Only a rotating subset is binding at any time, so the generator cannot over-fit a fixed discriminator; it must "keep running to stay in place" (Van Valen, 1973). In game-theoretic terms, alternating a learner against a moving set of critics drives it toward a point where no single sameness facet can be exploited without triggering another critic in a later rotation—an informal analogue of adversarial min–max training (Goodfellow et al., 2014).

Mutual-information view. Let R be the route and Y the surface text. Mechanism collapse corresponds to the pool's route distribution collapsing ($H(R) \to 0$) or to route becoming predictable from superficial features. In Regime II we can measure the analogous quantity across teachers: the mutual information $I(T; R)$ between teacher and route in the admitted pool. Low $I(T; R)$ with high joint entropy means neither teacher monopolizes any route.

Why it matters for consensus. Ensembles help only when their members' errors are decorrelated (Dietterich, 2000; Kuncheva and Whitaker, 2003). § 3.1 makes this precise for ACP: consensus purifies exactly the errors routes make independently. Mechanism plurality is therefore not cosmetic; it is the resource that the consensus gate consumes.

Counterfactual probes. A fraction of candidates is perturbed (sentence shuffles, negation flips) in the spirit of counterfactual augmentation (Kaushik et al., 2020), scored by all gates and critics, and counted as caught if any rejects them. Probes are never admitted or trained on; their catch rate audits whether the gates are sensitive to structural corruption.

What we observe on the logs. Fig. 10 shows the admitted Regime II lattice: admission rates per teacher×route cell differ by up to 3×, so the gates are selective about mechanism, yet every route retains 14–29% of the pool and the Qwen-teacher share stays between 49% and 61% in every round. In Regime I the admitted route mix is nearly uniform and stable over eight generations (route entropy 2.30 of 2.32 bits). Counterfactual probes are caught at 0.54–0.73 (Regime I setting means) and 0.48–0.55 (Regime II backbone means; Fig. 12).

Round route share ≤ 0.45 and cumulative share ≤ 0.60; 3 of 4 critics active, rotated every 3 rounds; reject on ≥ 2 votes; 15% counterfactual probes; β=2. Route caps and critic votes never rejected a candidate; critic rotations occurred on schedule and, in Regime II, on tripwire.

Teacher-divergence terms $D^{k,j}_{\text{route}} = \mathbb{E}_x[\text{KL}(P_{T_k}(\cdot \mid x) \,\|\, P_{T_j}(\cdot \mid x))_{\text{sym}}]$ with a floor; route-aware distillation (Hinton et al., 2015) with a diversity bonus on the student's route posterior; a route–template decoupling penalty; and retrieval-grounded routes, which our $\hat{\rho}$ analysis suggests are the most promising way to decorrelate knowledge errors.

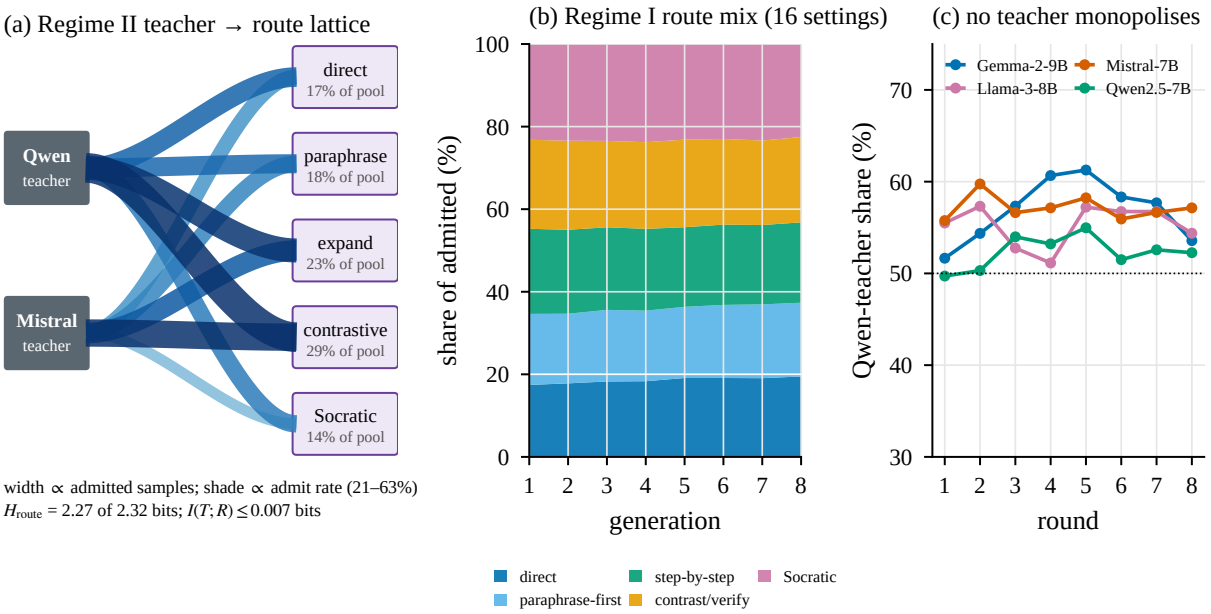


Figure 10: **T4 on real logs. (a)** Regime II teacher→route lattice of admitted samples, pooled over four backbones: edge width $\propto$ admitted samples, shade $\propto$ admission rate. **(b)** Regime I share of admitted samples per route over generations, averaged over all 16 settings. **(c)** Regime II share of admitted samples from the Qwen teacher per round.

A.7 Tripwires, adaptation and the gate interlock

ACP monitors a compact vector of collapse statistics, $\mathbf{m}(w) = [\text{admit rate}, H_{\text{route}}, \text{repetition}]_w$, and adapts when a threshold is crossed. A tripwire is a pre-specified, auditable trigger rather than a proof that collapse is impossible. The current adaptation is critic rotation in the next round; tightening floors, refreshing bands and reweighting clusters are natural further actions.

Signal	Threshold	Action	Firings observed
admit rate	$< 5\%$	rotate critics in $w{+}1$	none (I and II)
route entropy H_{route}	< 1 bit (max 2.32)	rotate critics in $w{+}1$	none (min. observed 2.22)
admitted bigram repetition	> 0.40 (Regime II)	rotate critics in $w{+}1$	Mistral-7B 8/8; others 1/8 (round 6)

Table 4: Tripwires as implemented, and how often they fired.

The interlock. Consensus → T1 → T2 → T3 → T4 is not a list of independent filters but a chain of dependencies: consensus makes admitted labels more correct; T1 keeps them learnable; T2 makes sure learnable data still carries rare support; T3 distributes that support across topics; T4 keeps the routes plural, which in turn is what lets consensus keep working in the next generation. Without T1, T2–T4 must fight head rote and junk simultaneously; without T4, consensus degenerates toward single-route majority voting.

B Experiments & Results

We assess whether ACP keeps recursive, label-free self-training stable—correct labels, learnable data, rare-token throughput, coverage and mechanism plurality—on a compute-feasible academic stack. All numbers and figures in this appendix are regenerated from the released per-round logs (§ B.7).

B.1 Implementation details

Component	Regime I (self-consuming tasks)	Regime II (open-web lattice)
Students	Gemma-2-9B-it, Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3, Qwen2.5-7B-Instruct (bf16)	Gemma-2-9B, Llama-3-8B, Mistral-7B-v0.3, Qwen2.5-7B (base, 4-bit NF4)
Generator	the current student (base + LoRA)	Qwen2.5-7B-Instruct, Mistral-7B-Instruct-v0.3 (4-bit)
Seeds per round	400 fresh training questions	40 FineWeb (sample-10BT) documents
Routes (K=5)	direct, paraphrase-first, step-by-step, contrast/verify, Socratic	direct continuation, paraphrase, expand, contrasting view, Socratic Q&A
Candidates per round	2,000	400 (2 teachers $\times$ 5 routes $\times$ 40)
Sampling	T=0.8, top-p 0.95	T=0.9, top-p 0.95
Consensus	$\geq$ 3 of 5 routes share the unique modal answer	cosine $\geq$ 0.55 cluster spanning $\geq$ 2 routes
T1 band	10th–90th pct. of H_{π_0} (post-consensus)	20th–80th pct. of H_{π_0} (all generated)
T2	new-3-gram rate $\geq$ 0.10; distinct-token ratio $\geq$ 0.20	tail-mass floor + new-3-gram rate $\geq$ 6% (or tail $\geq$ 0.5% and new $\geq$ 3%)
Dedup	MiniLM cosine $\geq$ 0.94, memory 4,000	MiniLM cosine $\geq$ 0.94, memory 2,000
T3	C=8 k-means clusters on seed prompts, KL $\leq$ 1.5	same
T4	round share $\leq$ 0.45, cumulative $\leq$ 0.60; 3 of 4 critics active, rotated every 3 rounds; reject on $\geq$ 2 votes; 15% counterfactual probes	same
Tripwires	admit rate $<$ 5% or $H_{\text{route}} <$ 1 bit	additionally admitted repetition $>$ 0.40
LoRA	r=16, α=32, dropout 0.05; q,k,v,o,gate,up,down	r=16, α=32, dropout 0.05; q,k,v,o
Optimisation	AdamW, lr 10^{-4}, eff. batch 16, $\leq$ 64 steps/round, max len 1024	AdamW, lr 10^{-4}, eff. batch 16, $\leq$ 50 steps/round, max len 512
Training data	cumulative admitted pool, route-aware sampling (β=2)	same
Evaluation	greedy, full split (GSM8K 1,319; ARC-C 1,172; CSQA 1,221; BoolQ 3,270)	data-health diagnostics
Hardware	1$\times$H100 80GB, vLLM (Kwon et al., 2023)	2$\times$T4 16GB (Kaggle), $\approx$5 h per backbone

Table 5: Full configuration of both regimes. The frozen reference π_0 for T1 is the student with LoRA adapters disabled, so no second model copy is needed.

Answer extraction. GSM8K answers are parsed from `Answer:`, `\boxed{}` or the last number; multiple-choice letters and yes/no from the `Answer:` line. Truncated generations are excluded from consensus. Format compliance was 100% at every evaluation step.

Critics. *Template*: refusal/boilerplate openings or $\geq 2/3$ repeated lines. *Style*: coefficient of variation of sentence length < 0.08 over ≥ 5 sentences. *Entropy shape*: $H_{\pi_0} < 0.15$ bits/token. *Lexical diversity*: distinct-token ratio < 0.30.

Counterfactual probes. With probability 0.15 a candidate is copied and perturbed by shuffling its sentences or inserting a negation after the first copula. Probes are scored by all gates and critics and counted as *caught* if any rejects them. They are never admitted or trained on.

B.2 Proofs and derivations

Proof of Proposition 1. Fix a prompt and let each of the K routes independently output the correct answer with probability p and each of L wrong answers with probability $(1-p)/L$. The consensus gate admits the correct answer only if it receives $\geq m$ votes, so $\Pr[\text{admit correct}] \leq A(p) = \bar{F}_K(m; p)$.

Admitting a wrong answer requires that *some* wrong answer ℓ receive $\geq m$ votes. Marginally, the vote count of ℓ is $\mathrm{Bin}(K, (1-p)/L)$, so by the union bound $\Pr[\text{admit wrong}] \leq B(p) = L\,\bar{F}_K(m; (1-p)/L)$.

For the lower bound on precision we also need $\Pr[\text{admit correct}]$ from below. With $m > K/2$ (here m=3, K=5), at most one answer can reach m votes, so the mode is automatically unique whenever any answer reaches m, and $\Pr[\text{admit correct}] = A(p)$ exactly. Precision is $\pi_\perp = \Pr[\text{admit correct}]/(\Pr[\text{admit correct}] + \Pr[\text{admit wrong}]) \geq A/(A+B)$. As $L \to \infty$, $(1-p)/L \to 0$ and $L\,\bar{F}_K(m; (1-p)/L) = O(L^{1-m}) \to 0$ for $m \geq 2$, hence $\pi_\perp \to 1$. □

Common-mode model (Eq. (5)). With probability ρ a prompt is common-mode: all K routes emit one shared answer, which is correct with probability p and always reaches consensus. With probability $1-\rho$ routes behave independently as above. Admitted-correct mass is $\rho p + (1-\rho)A$ and admitted-wrong mass is $\rho(1-p) + (1-\rho)B$; their ratio gives Eq. (5). Because π_ρ is a ratio of affine functions of ρ, it is monotone in ρ, and inverting at the observed precision π gives

$$\hat{\rho} = \frac{A - \pi(A+B)}{\pi - \pi(A+B) - p + A}.$$

Tab. 6 reports the inputs and estimates. ARC-C items occasionally have five options; using L=3 is slightly conservative there.

Student	Task	L	p (gen.)	$\pi_\perp$ bound	observed π	gap	$\hat{\rho}$
Gemma-2-9B	GSM8K	∞	0.889	1.000	0.958	+0.042	0.38
Gemma-2-9B	ARC-C	3	0.917	0.999	0.945	+0.054	0.66
Gemma-2-9B	CSQA	4	0.757	0.991	0.801	+0.190	0.80
Gemma-2-9B	BoolQ	1	0.859	0.978	0.891	+0.087	0.73
Llama-3.1-8B	GSM8K	∞	0.874	1.000	0.989	+0.011	0.09
Llama-3.1-8B	ARC-C	3	0.817	0.994	0.929	+0.065	0.36
Llama-3.1-8B	CSQA	4	0.720	0.986	0.802	+0.183	0.66
Llama-3.1-8B	BoolQ	1	0.738	0.884	0.922	-0.038	-0.26
Mistral-7B	GSM8K	∞	0.547	1.000	0.916	+0.084	0.12
Mistral-7B	ARC-C	3	0.745	0.982	0.860	+0.122	0.49
Mistral-7B	CSQA	4	0.665	0.975	0.763	+0.211	0.64
Mistral-7B	BoolQ	1	0.826	0.960	0.861	+0.099	0.74
Qwen2.5-7B	GSM8K	∞	0.923	1.000	0.974	+0.026	0.33
Qwen2.5-7B	ARC-C	3	0.897	0.999	0.933	+0.066	0.64
Qwen2.5-7B	CSQA	4	0.802	0.995	0.842	+0.153	0.78
Qwen2.5-7B	BoolQ	1	0.832	0.964	0.863	+0.101	0.76

Table 6: Consensus-precision bound under independence ($\pi_\perp$, Eq. (4)) vs. observed admitted precision, and the implied common-mode fraction $\hat{\rho}$. The ordering GSM8K < ARC-C < CSQA holds for every student. Llama/BoolQ is the one case where observed precision exceeds the independence prediction ($\hat{\rho} < 0$), because precision is measured after all gates and T1/dedup remove further errors.

Why the lift anticorrelates with generator quality. Write the precision lift as $\Lambda(p) = \pi_\rho(p) - p$. Under pure independence with large L, $\Lambda \approx 1 - p$, decreasing in p. Under the common-mode model, Λ is scaled by the independent fraction $(1-\rho)$ and still vanishes as $p \to 1$. Both predict the negative relation observed across 128 rounds ($r = -0.75$). The strength of the empirical relation ($|r|$ well below 1) is itself explained by ρ varying across tasks.

B.3 Extended Regime I results

How generator erosion is measured. For each setting we fit a least-squares line to the generated-label precision over rounds 1–8 and report its total change (slope × 7). The trend is negative in 14/16 settings (mean −2.6 pp, minimum −7.7 pp for Llama/BoolQ). The same statistic on admitted precision averages −0.4 pp, so ACP absorbs $1 - 0.4/2.6 \approx 83\%$ of the erosion. Endpoint differences (round 8 minus round 1) are noisier and give a smaller mean erosion (−1.4 pp, negative in 12/16), with admitted precision again flat or rising; Tab. 8 lists the endpoints.

Student	Task	G0	G1	G2	G3	G4	G5	G6	G7	G8
Gemma-2-9B	GSM8K	88.9	90.0	88.6	89.1	89.2	88.3	88.9	88.9	87.9
Gemma-2-9B	ARC-C	91.7	91.4	92.0	90.5	91.6	91.4	92.1	91.7	91.0
Gemma-2-9B	CSQA	79.0	79.5	80.7	79.4	80.1	79.6	79.4	80.5	79.8
Gemma-2-9B	BoolQ	88.1	87.5	87.0	86.8	87.5	86.4	86.4	87.2	87.2
Llama-3.1-8B	GSM8K	85.9	86.2	85.4	86.2	85.7	84.4	85.1	84.6	84.3
Llama-3.1-8B	ARC-C	87.2	86.8	85.3	85.2	85.0	84.0	82.8	82.4	83.6
Llama-3.1-8B	CSQA	77.4	77.7	78.4	78.5	77.3	78.5	76.9	77.9	77.4
Llama-3.1-8B	BoolQ	81.9	80.7	80.6	81.7	80.4	81.0	81.0	82.7	81.2
Mistral-7B	GSM8K	56.8	55.0	57.2	58.5	58.0	56.4	58.2	58.3	56.7
Mistral-7B	ARC-C	77.0	79.0	77.5	77.0	78.7	77.6	76.1	75.9	75.3
Mistral-7B	CSQA	70.5	69.9	71.1	70.4	71.8	71.6	70.5	71.9	71.3
Mistral-7B	BoolQ	83.5	83.1	83.4	82.2	82.0	82.4	82.8	83.6	83.5
Qwen2.5-7B	GSM8K	90.3	91.4	89.6	88.8	89.8	89.2	89.8	90.8	90.1
Qwen2.5-7B	ARC-C	91.5	91.4	90.5	89.9	91.5	91.5	90.0	89.8	89.0
Qwen2.5-7B	CSQA	82.2	80.9	82.8	81.3	81.7	80.6	80.4	80.3	81.3
Qwen2.5-7B	BoolQ	84.3	83.8	84.5	84.8	85.0	84.3	85.1	85.7	85.2
Mistral-7B (naive)	GSM8K	56.8	56.9	57.9	57.9	58.2	–	–	–	–

Table 7: Accuracy (%) after every generation for all 16 settings, plus the four-generation naive loop.

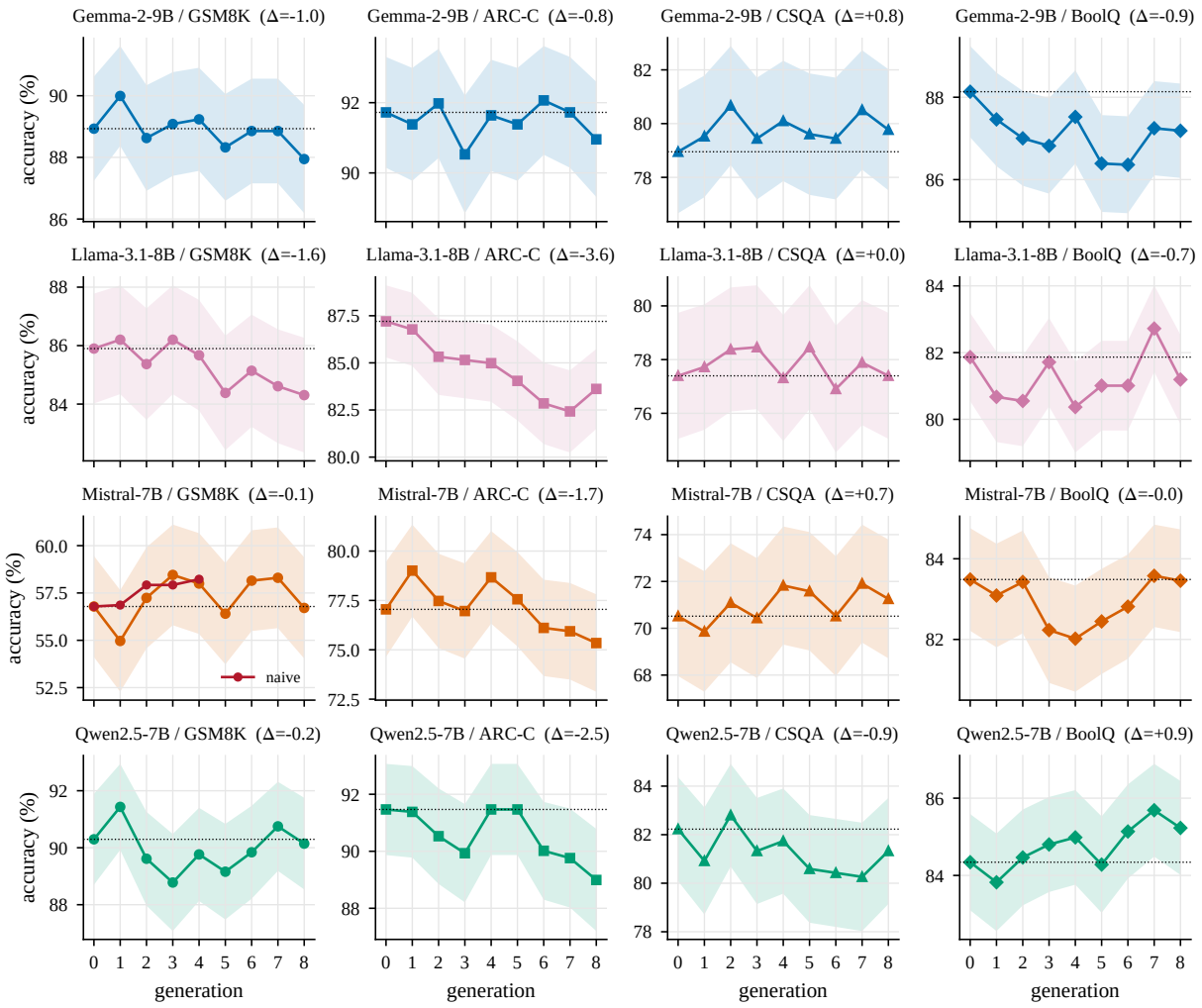


Figure 11: Per-setting accuracy trajectories with 95% binomial confidence bands (shaded) and the base-model accuracy (dotted). The naive loop is overlaid for Mistral/GSM8K. All trajectories except Llama/ARC-C remain inside or near the base model's confidence band.

Student	Task	consensus	T1	T2	dedup	T3/T4	admitted	H_{route}	CF catch	gen. prec. R1→R8
Gemma-2-9B	GSM8K	1103	180	2	198	0	517	2.289	0.69	92.9→90.0
Gemma-2-9B	ARC-C	400	320	60	46	0	1172	2.290	0.65	91.7→90.9
Gemma-2-9B	CSQA	191	362	0	28	0	1418	2.302	0.66	76.1→77.0
Gemma-2-9B	BoolQ	137	373	10	102	0	1379	2.307	0.68	86.2→85.8
Llama-3.1-8B	GSM8K	297	341	8	289	0	1066	2.314	0.70	88.4→87.8
Llama-3.1-8B	ARC-C	329	335	66	44	0	1226	2.316	0.70	84.6→78.8
Llama-3.1-8B	CSQA	328	335	0	14	0	1322	2.312	0.67	72.0→72.0
Llama-3.1-8B	BoolQ	714	258	4	63	0	961	2.308	0.73	79.0→73.7
Mistral-7B	GSM8K	999	201	30	92	0	679	2.314	0.73	52.8→55.7
Mistral-7B	ARC-C	400	320	83	41	0	1155	2.303	0.69	77.0→74.4
Mistral-7B	CSQA	404	320	0	14	0	1262	2.306	0.70	69.1→67.7
Mistral-7B	BoolQ	199	362	9	202	0	1229	2.312	0.72	83.5→84.2
Qwen2.5-7B	GSM8K	139	373	54	260	0	1174	2.300	0.66	95.0→92.5
Qwen2.5-7B	ARC-C	130	374	111	74	0	1311	2.296	0.61	89.8→87.3
Qwen2.5-7B	CSQA	176	366	0	34	0	1424	2.291	0.54	80.5→81.5
Qwen2.5-7B	BoolQ	152	370	9	108	0	1361	2.305	0.69	85.2→82.9

Table 8: Mean per-round admission funnel (of 2,000 candidates), admitted route entropy (maximum $\log_2 5$=2.32 bits), counterfactual-probe catch rate, and the precision of the student's own generations in round 1 vs. round 8. The T3/T4 caps and critic vote threshold never rejected a candidate; tripwires never fired in Regime I.

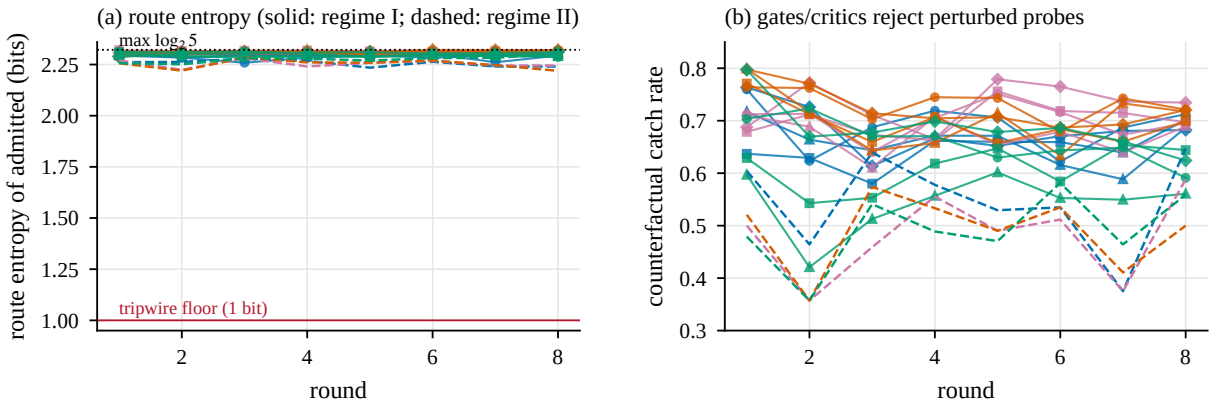


Figure 12: **(a)** Admitted route entropy stays within 3% of the maximum in both regimes, far above the 1-bit tripwire. **(b)** Counterfactual probes (sentence shuffles and negation flips) are rejected at 42–80%; Regime II (dashed) is lower because open text tolerates reordering better than structured answers.

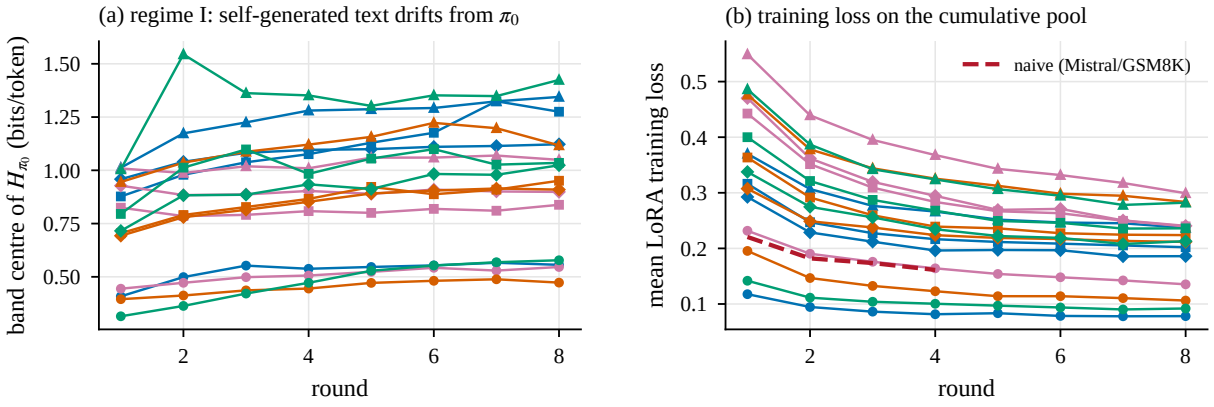


Figure 13: **(a)** The centre of the T1 band rises in every setting: the student's own outputs become steadily more surprising to the frozen base model, a direct measurement of self-consumption drift. **(b)** Mean LoRA training loss on the cumulative pool decreases smoothly; the naive pool (dashed) is consistently harder to fit, consistent with its 48% label noise.

Naive vs. ACP, in numbers (Mistral/GSM8K). Generations 1–4 admit 2,000 / 2,000 / 2,000 / 2,000 naive samples with label precision 52.9 / 53.3 / 49.9 / 51.6%, versus 670 / 651 / 672 / 742 ACP samples at 90.4 / 94.0 / 90.2 / 91.4%. Cumulatively, the naive pool reaches 8,000 samples with 3,849 wrong (48.1%), while the ACP pool reaches 2,735 with 233 wrong (8.5%). Accuracy at G4 is 58.2% (naive) vs. 58.0% (ACP), well within the ±2.7 pp confidence half-width.

B.4 Per-round screening outcomes

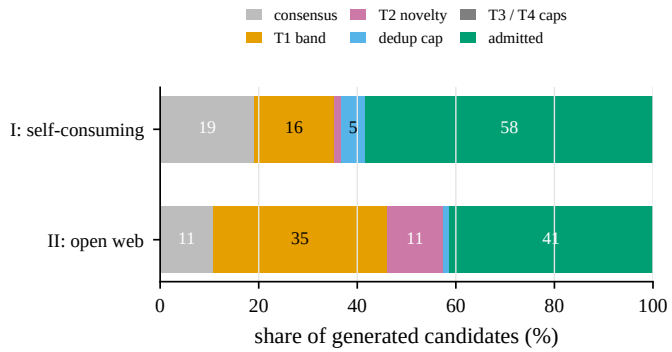


Figure 14: **Where candidates are rejected** (mean share of generated candidates). Consensus and T1 do most of the work; T2 matters for open text; the T3/T4 caps never bound in our runs.

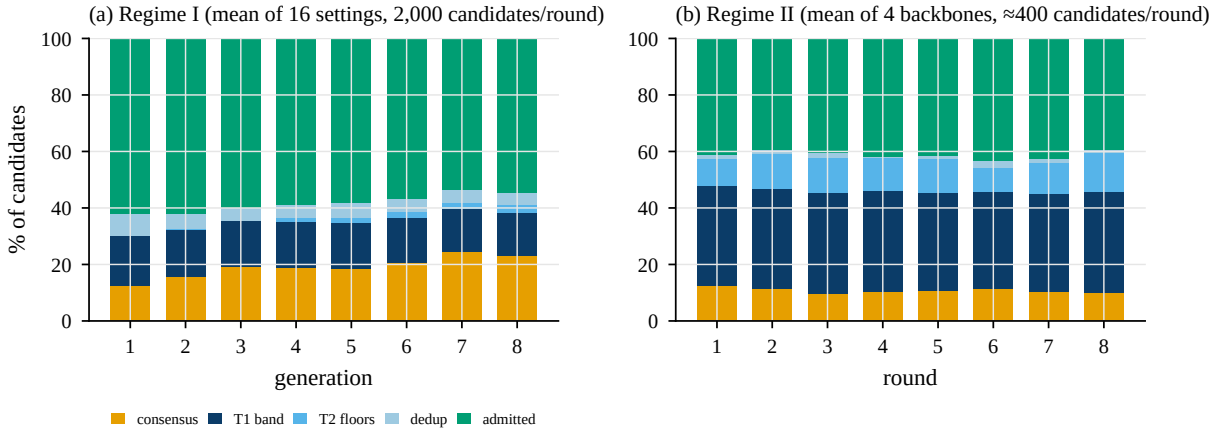


Figure 15: **Screening and data-health dashboard per round.** Stacked bars show where generated candidates go in each round. **(a)** Regime I, mean over 16 settings: the consensus share grows over generations (from 13% in round 1 to 23% in round 8), mirroring the erosion of the student's own label precision, while the T1 share stays near its quantile-defined level. **(b)** Regime II, mean over four backbones: T1 and T2 dominate and are stable across rounds. T3 and T4 never rejected a candidate and are therefore absent.

B.5 Extended Regime II results and the repetition audit

Size-matched repetition. Bigram repetition, $\sum_b (c_b - 1)\mathbb{1}[c_b > 1]/(N-1)$ over bigram counts c_b, increases with corpus size N. Admitted pools contain ≈40k tokens per round, generated pools ≈89k. For the Qwen2.5-7B run, where the exact tokenizer is public, we recomputed the metric (reproducing the logged values exactly) on random within-round subsamples at 25, 50, 75 and 100% of the admitted pool. We fit repetition $= a \log N + b$ (slope $a \approx 0.10$ per nat in every round) and extrapolated to the generated pool's size. Within-round subsampling keeps the prompt set fixed; random cross-round pooling would instead mix in more prompts and *understate* repetition. Tab. 11 shows that the raw gap of 0.092 shrinks to 0.014 after size matching: 85% of the apparent improvement is an artifact, and the remainder is positive in every round.

	Gemma	Llama	Mistral	Qwen
admitted / generated	37.9%	43.1%	42.3%	42.0%
tail mass (adm./gen.)	1.31×	1.16×	1.13×	1.17×
rounds improved	8/8	8/8	7/8	8/8
H_{route} (bits, max 2.32)	2.26	2.25	2.25	2.28
$H(T,R)/H_{\max}$	97.7%	97.8%	97.6%	98.5%
$I(T;R)$ (bits)	.006	.005	.004	.007
CF probe catch rate	.55	.48	.49	.49
tripwire firings	1/8	1/8	8/8	1/8

Table 9: **Regime II summary.** $H(T,R)$: joint teacher–route entropy of the admitted pool; $I(T;R)$: teacher–route mutual information.

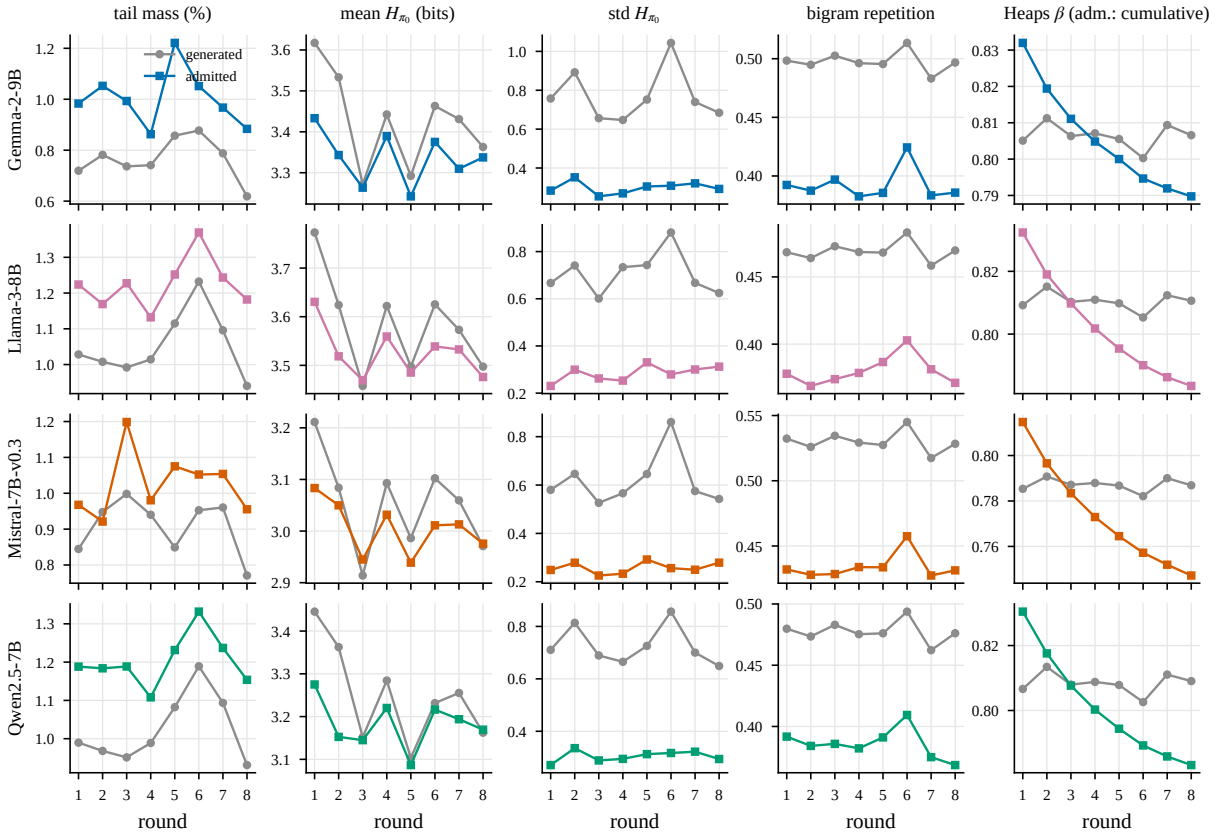


Figure 16: All Regime II diagnostics per backbone and round (grey: generated; colour: admitted). The admitted Heaps β is logged on the *cumulative* pool, whose size grows each round, so its decline reflects sub-linear vocabulary growth (Heaps' law) rather than a loss of diversity; it is not comparable to the per-round generated value.

Backbone	Quantity	R1	R2	R3	R4	R5	R6	R7	R8
Gemma-2-9B	admitted	151	149	150	150	142	168	156	140
	T1 rejects	141	144	143	143	138	135	142	143
	T2 rejects	53	60	61	61	72	42	56	72
	tail gen.	0.72	0.78	0.74	0.74	0.86	0.88	0.79	0.62
	tail adm.	0.98	1.05	0.99	0.86	1.22	1.05	0.97	0.88
	rep. adm.	0.392	0.387	0.397	0.383	0.386	0.424	0.383	0.386
	H_{route}	2.26	2.26	2.28	2.27	2.23	2.26	2.24	2.24
Llama-3-8B	admitted	173	164	163	176	180	178	178	160
	T1 rejects	143	143	141	136	135	135	139	143
	T2 rejects	28	42	46	42	37	29	36	51
	tail gen.	1.03	1.01	0.99	1.01	1.11	1.23	1.10	0.94
	tail adm.	1.22	1.17	1.23	1.13	1.25	1.37	1.24	1.18
	rep. adm.	0.378	0.369	0.374	0.379	0.387	0.403	0.381	0.371
	H_{route}	2.27	2.22	2.28	2.24	2.26	2.27	2.25	2.25
Mistral-7B-v0.3	admitted	165	159	166	168	170	177	173	168
	T1 rejects	138	141	138	144	139	135	138	139
	T2 rejects	45	48	46	42	44	30	40	43
	tail gen.	0.84	0.95	1.00	0.94	0.85	0.95	0.96	0.77
	tail adm.	0.97	0.92	1.20	0.98	1.08	1.05	1.05	0.96
	rep. adm.	0.432	0.428	0.428	0.434	0.434	0.457	0.427	0.431
	H_{route}	2.26	2.22	2.28	2.26	2.26	2.27	2.25	2.22
Qwen2.5-7B	admitted	173	161	163	171	171	167	175	155
	T1 rejects	142	142	143	141	139	142	138	144
	T2 rejects	28	46	45	42	40	35	41	53
	tail gen.	0.99	0.97	0.95	0.99	1.08	1.19	1.09	0.93
	tail adm.	1.19	1.18	1.19	1.11	1.23	1.33	1.24	1.15
	rep. adm.	0.392	0.384	0.386	0.382	0.391	0.409	0.375	0.368
	H_{route}	2.26	2.25	2.28	2.28	2.27	2.28	2.28	2.30

Table 10: Per-round Regime II diagnostics. Candidate pools are identical across backbones (seeded teachers), so differences arise only from each student's frozen reference π_0 and tokenizer.

Round	1	2	3	4	5	6	7	8
generated	0.480	0.473	0.483	0.475	0.476	0.494	0.462	0.476
admitted (raw)	0.392	0.384	0.386	0.382	0.391	0.409	0.375	0.368
admitted (size-matched)	0.468	0.459	0.469	0.454	0.469	0.492	0.447	0.452
residual gap	0.012	0.015	0.014	0.022	0.007	0.002	0.016	0.024

Table 11: Repetition audit for Qwen2.5-7B (Regime II).

B.6 Pilot: reduced ACP across languages and backbones

Overview and motivation. Before the experiments in the main paper, we ran an earlier, reduced version of the pipeline to probe two generalization axes: *backbone* (Llama-3-8B and Mistral-7B-v0.3 students) and *language* (English FineWeb, Spanish and Hindi FineWeb-2). It used T1+T2+T3 only—no consensus gate and no Red-Queen critics—with a single Qwen2.5-7B-Instruct teacher (8-bit) and two generation modes per prompt (plain continuation and language-matched chain-of-thought), 8-bit students, learning rate 7×10^{-6}, five rounds, and `paraphrase-multilingual-MiniLM-L12-v2` embeddings for non-English runs. We report it as supporting evidence that the data-health effects of the gates transfer across languages and scripts; because its configuration differs from the main experiments, it is not directly comparable to them.

Run	Student	Data	Teacher	Gates	Rounds
EN-L	Llama-3-8B	FineWeb (English)	Qwen2.5-7B-I (8-bit)	T1+T2+T3	5
EN-M	Mistral-7B-v0.3	FineWeb (English)	Qwen2.5-7B-I (8-bit)	T1+T2+T3	5
ES-L	Llama-3-8B	FineWeb-2 (Spanish)	Qwen2.5-7B-I (8-bit)	T1+T2+T3	5
ES-M	Mistral-7B-v0.3	FineWeb-2 (Spanish)	Qwen2.5-7B-I (8-bit)	T1+T2+T3	5
HI-L	Llama-3-8B	FineWeb-2 (Hindi)	Qwen2.5-7B-I (8-bit)	T1+T2+T3	5

Table 12: Configurations of the five reduced-ACP pilot runs. All share LoRA r=16, α=32 on q,k,v,o and one epoch per round on the cumulative accepted pool.

Observations. Across all five runs, admitted pools show lower repetition and tighter entropy dispersion than generated pools in every round, and no run shows the classical cascade of tail death, repetition explosion or entropy collapse. Admitted tail mass is at or above generated tail mass in most rounds, with the largest margin for Hindi. Spanish shows the steepest rare-token decline in both pools, suggesting that tail floors should be calibrated per language. Hindi required lower entropy-band thresholds, since English-calibrated bands over-reject Devanagari text. Two caveats apply: the repetition gap is partly a pool-size artifact (§ B.5), and the admitted Heaps β is computed on a growing pool. The pilot therefore supports transfer of the *data-health* effects, not of the label-purification results, which require the consensus gate.

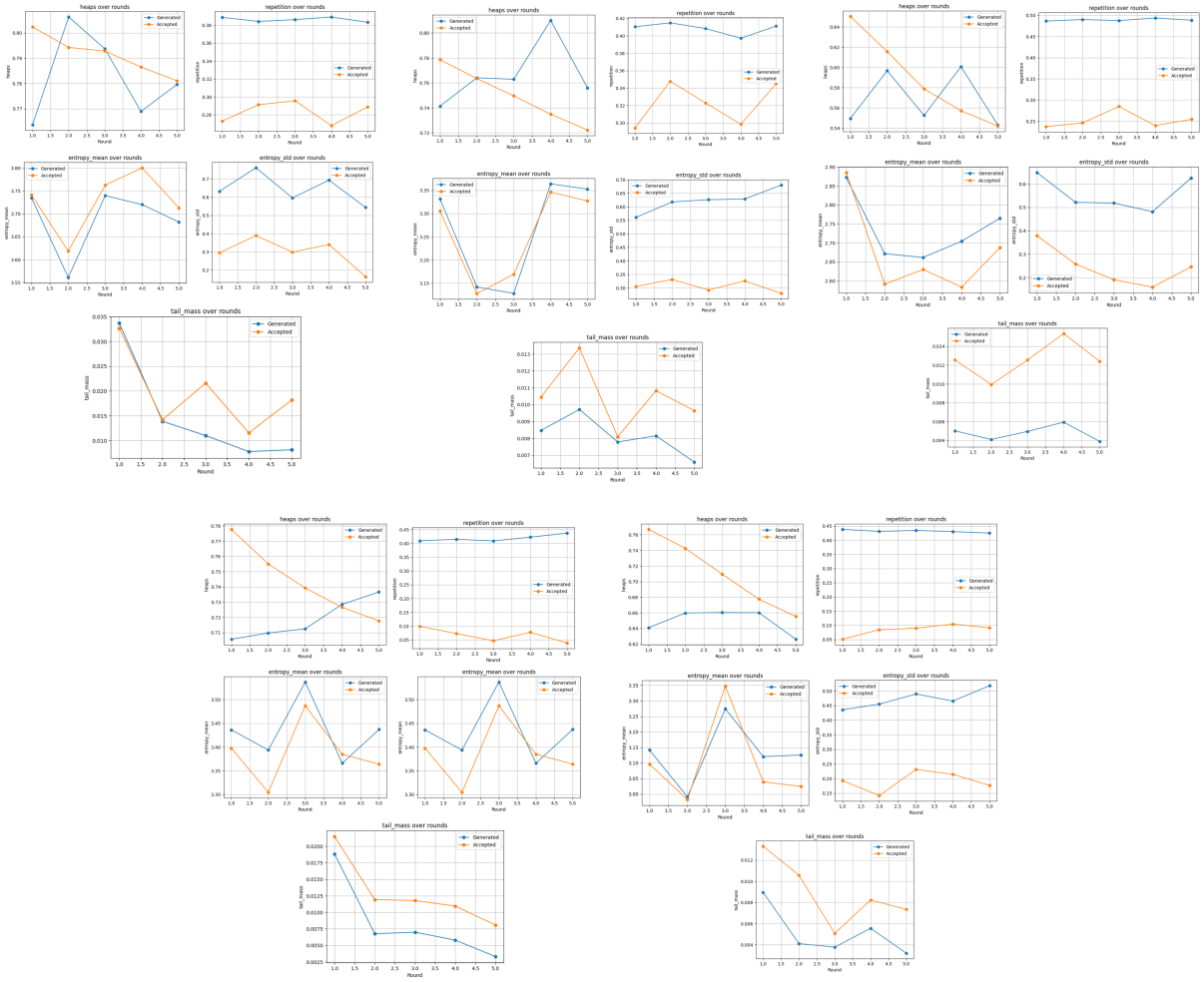


Figure 17: Reduced-ACP pilot (blue: generated; orange: admitted). Top: Llama-3-8B English, Mistral-7B English, Llama-3-8B Hindi. Bottom: Llama-3-8B Spanish, Mistral-7B Spanish. Panels per run: Heaps β, repetition, entropy mean, entropy s.d., tail mass.

B.7 Reproducibility checklist and compute

- **Code and logs.** Experiment code (`code/`), per-round statistics for every run (`raw_logs/`), and analysis scripts (`analysis/`) are released. `expB.py` (Regime I statistics), `expA.py` (Regime II statistics), `consensus_bound.py` (Proposition 1, $\hat{\rho}$), `repetition_audit.py` (§ B.5), `make_figures.py` and `make_figures_v2.py` regenerate every number, table and figure.
- **Seeds.** Fixed (42) throughout; teacher generations in Regime II are therefore shared across backbones.
- **Gold-label isolation.** Gold labels are read only by the evaluation routine and by the logging of label precision, both after admission decisions are final.
- **Statistics.** Two-proportion tests (unpaired, conservative) with Bonferroni correction over 16 settings; Pearson correlations at the round level.
- **Compute.** Regime I: one H100 80GB; each phase (evaluation, generation, filtering, training) runs in a fresh subprocess and is resumable from per-round checkpoints; filtering takes ≈35 s and training ≈52 s per round. Regime II: Kaggle's free 2×T4 tier, ≈5 h per backbone.
- **Provenance of figures.** Fig. 1 is reproduced from Villalobos et al. (2024). The pilot figures (Fig. 17) come from the earlier reduced-ACP runs. Every other figure is generated from the logs in this archive.